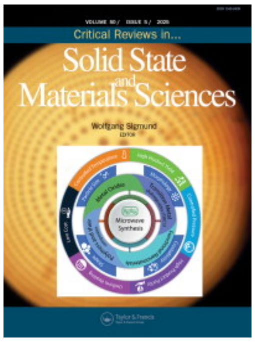

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REVIEW ARTICLE

# A critical review of consolidation processes for ODS steels: Development as clad tube materials in Gen-IV nuclear reactors

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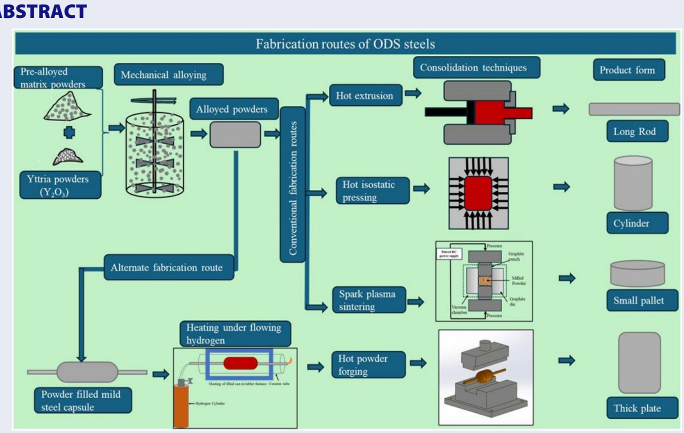
**ABSTRACT**

Oxide Dispersion Strengthened (ODS) steels have been developed over the past decades as structural materials for fission reactors. In recent years, significant interest has grown in their use as cladding tubes in next-generation Generation-IV (Gen-IV) nuclear reactors. Cladding tubes serve to separate reactor fuel from the coolant and are subjected to extremely high temperatures, radiation doses, and biaxial stresses. As such, they must exhibit excellent high-temperature strength, radiation resistance, and creep resistance. To meet these demanding requirements, two primary grades, ferritic-martensitic (F/M) and ferritic ODS steels, are being extensively studied and developed for use as clad tube materials in Gen-IV nuclear reactors. ODS steels are mainly prepared through powder metallurgy (PM) routes, in which elemental powders or pre-alloyed powders are mechanically alloyed with nanosized yttria powders in a high-energy attritor or ball mill. The mechanically alloyed powders are consolidated through different routes, the main ones being hot extrusion (HE), hot isostatic pressing (HIP), and spark plasma sintering (SPS), with hot powder forging (HPF) being recently developed as an alternative consolidation route. The limitations associated with these conventional routes and opportunities with alternate consolidation routes are adequately discussed. The influence of various potential alloying elements on the microstructure and mechanical properties of ODS steels is widely reviewed here. With careful selection of compositions and processing routes, alloys with up to 1.5 wt.% yttria can be fabricated. Higher yttria contents lead to ODS steels with improved elevated temperature strength and creep resistance. Anisotropy in ODS steels with high Cr contents (>11 wt.%) is a concern, but it can be overcome by the application of the newly developed powder forging route; furthermore, ODS steels consolidated through the HPF route develop high density of the product. Finally, this review identifies potential candidate cladding materials for the upcoming Gen-IV nuclear reactors.

**KEYWORDS**
ODS steels; clad tubes; powder metallurgy; consolidation routes; Gen-IV nuclear reactors

**GRAPHICAL ABSTRACT**

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## Table of contents



## 1. Introduction

Concerns about global warming because of greenhouse gas emissions are increasing, and time is running out to reduce global emissions and avoid serious climate change consequences.[1] This urgency necessitates the use of all low-carbon energy generation solutions to move away from fossil fuels. Even though renewable energies are essential, nuclear energy will definitely play a role, as renewable energies should be accompanied by a sustainable source of energy. Nuclear power plants in 32 nations are already cutting $CO_2$ emissions in the global power industry by roughly 10%,[2] and 19 countries are actively building over 50 additional power reactors with a capacity of around 54 GW.[3] Nuclear power is one of the few low-carbon energy sources that can provide electricity, heat, and hydrogen. Many new nuclear technologies, such as small modular reactors (SMRs) and advanced nuclear reactors, are opening up new possibilities.[3]

Due to their low thermal efficiency, currently operating nuclear reactor plants worldwide will be unable to fulfill the growing power demand, as they would be unable to withstand the extremely high temperatures and radiation doses required for high fuel burn-up to increase the plant's thermal efficiency. Advanced (Gen-IV) nuclear reactors (fission) are being developed for more power generation with minimal wastage and a longer reactor life. The operating temperatures for these reactors are from 500 to 1000 °C and can withstand intense neutron flux of up to 200 dpa (displacement per atom).[4] After the year 2030, the majority of the Gen-IV technologies are expected to be available for commercial development. The effective operation of nuclear reactors is mainly dependent on structural materials (in-core and out-core components), which are directly exposed to severe neutron irradiation and high temperatures.[5] The successful functioning of Gen-IV nuclear reactors is mainly dependent on the performance of structural materials. The structural materials are primarily categorized into two main parts. The first one comprises the non-replaceable (long-lasting) components that surround and support the core; the second one includes the core components that contain fuel themselves. These are replaceable components once the fuel has reached the desired burn-up rate or when failure is expected. If the performance of core structural components such as cladding tubes and wrapper/duct materials is satisfactory to exposure to very high temperatures, then the high burn-up of the fuel can be achieved in the nuclear reactor.

The aim of this review is to explain the developments employed in ODS steels to be used as cladding materials for advanced nuclear power plants. This starts with the essential need for nuclear energy and the various types and generations (from Gen-I to Gen-IV) of nuclear power plants. Further, a comprehensive review of the materials required for Gen-IV has been done. The development of ODS steels and the effect of various alloying elements in their composition are discussed. Various consolidation routes along with their limitations and the development of new fabrication routes for ODS steels are comprehensively presented. A systematic review of the limitations associated with conventional consolidation processes and opportunities linked to newly developed hot powder forging routes has not been previously attempted. The correlation between microstructure and mechanical properties of ODS steels produced by various fabrication routes is widely discussed. This review paper discusses the current issues with the production of ODS steels and presents recent developments to meet the requirements for Gen-IV advanced nuclear reactors.

### 1.1. Nuclear as a green energy

The distribution of energy production due to different sources is presented in Figure 1. It can be seen that most of the world's energy is produced by burning fossil fuels such as coal. Coal is the main fossil fuel that produces extensive $CO_2$ emission per unit of energy produced. The main source of emissions from coal usage is in the generation of electricity, with coal-fired electric power generators being among the largest emitters of $CO_2$ per kilowatt-hour (kWh) of energy produced. Coal combustion is responsible for almost 40% of global $CO_2$ emissions from the energy sector, as well as significant local air pollution, which causes millions of premature deaths annually.[7,8] As a major source of global air pollution, it emphasizes the urgent need to shift to green energy resources. Nuclear power plants can be a suitable replacement for coal-fired power plants, as they are both thermal power plants that rely on similar components. Simply replacing 1% of coal-generated electricity with nuclear power would cut annual emissions by roughly 100 Mt $CO_2$.[3]

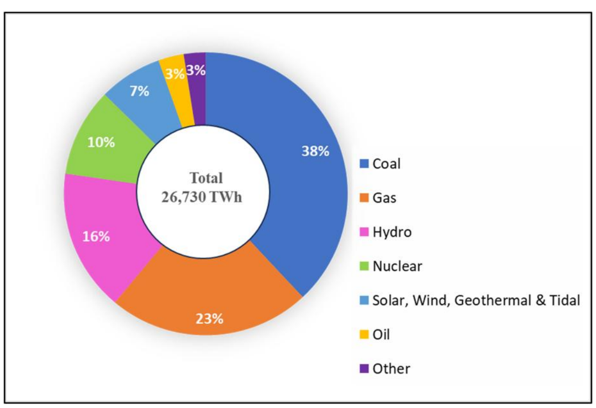


**Figure 1.** World electricity production by source in 2018 (Adapted from[6]).

The world would need to generate even more power to meet the growing demand for electricity for economic growth. The increase in worldwide demand for energy expected in the twenty-first century has spurred international cooperation to consider ways to meet future energy needs while maintaining and improving the environment. Since the year 2000, the majority of advanced nuclear systems have been developed under the aegis of the international research and development community, namely the Generation IV (Gen-IV) Initiative Forum (GIF), in order to meet the world's energy demands with safety, sustainability, low cost, and minimal nuclear waste. Nuclear power makes a significant contribution to global electricity generation, providing 10% of the global electricity supply in 2018.[6] Electricity generation through various sources is shown in the pie chart in Figure 1. As of Dec 2020, there were 441 nuclear power reactors in operation in 32 countries around the world, with a combined capacity of about 392 GW).[9] Nuclear power plays a much bigger role in advanced economies, where it makes up to 18% of total electricity generation with the lowest $CO_2$ emission.[2]

### 1.2. Various generations of nuclear fission reactors

Nuclear fission reactors are classified according to their generation; these are Gen-I, II, III, III+, and IV. Generation I (Gen-I) reactors are those that were designed in the 1950s and 1960s. Gen-I pertains to the prototype and power reactors which have been deployed for commercial electricity generation. These reactors were primarily designed and constructed for feasible electric power generation through nuclear energy. The understanding acquired from Gen-I reactors was used to develop and build larger reactors for commercial power generation.

Gen-II reactors belong to the class of commercial reactors that were developed and designed to be cost-effective and reliable. These (Gen-II) reactors were among the first to utilize nuclear energy for power generation on a large scale. The first commercial Gen-II reactors were constructed in the 1970s and were typically designed for 40 years of operational lifetime.[10] Currently, most of the nuclear power plants in this category are in operation around the world. These are pressurized water reactor (PWR), boiling water reactor (BWR), advanced gas cooled reactor (AGR), and CANada Deuterium Uranium Reactors (CANDU).[10] These reactors are mainly referred to as light weight reactors (LWRs). Gen-II reactors utilize standard safety measures such as electrical or mechanical operations that are automatically initiated and, in many instances, can be initiated manually.

Gen-III reactors are in the category of advanced versions of Gen-II reactors with enhanced design and superior safety measurements. The advancement in fuel technology and reactor design makes them have greater thermal efficiency, which safely reduces the overall capital as well as maintenance costs. With improved technology in Gen-III, they were planned for an operational life of 60 years or more. These reactors were constructed in 1990 and remain operational till now. An evolutionary development in the design of Gen-III reactors is called Gen-III+ reactors, which offer significant safety features over Gen-III reactors and are in an advanced stage of construction for commercial use.

Currently, 18% of the world's electricity is fulfilled by nuclear fission reactors. The global distribution of 441 nuclear reactors is aging, and they will need to be replaced and upgraded to keep up with and assume a larger share of the increasing global electricity demand. To fulfill the world's increasing electricity demands, new-generation nuclear reactors are being focused on by the international nuclear research and development community. In the year 2000, a new program known as the Generation-IV International Forum (GIF) was launched with the joint effort of ten countries (Argentina, Brazil, Canada, France, Japan, Korea, South Africa, Switzerland, the United Kingdom, and the United States) in collaboration with the International Atomic Energy Agency (IAEA) and the OECD Nuclear Energy Agency (NEA).[11] The main objective of this initiative is to expand the use of nuclear energy by developing further advancements in nuclear energy system design, such as improving resource use, safety features, efficiency, sustainability, economics-profitability, and useful reactor life with the least amount of production waste. These systems are expected to be deployed globally by 2030 (Figure 2a).

To achieve Gen-IV goals, six designs have been chosen for future research, development, and deployment from a few other available reactors. Three of them are fast reactors, such as (1) sodium-cooled fast reactor (SFR), (2) gas-cooled fast reactor (GFR), (3) lead-cooled fast reactor (LFR), and the other three are based on thermal reactors, i.e., (4) very-high-temperature reactor (VHTR), (5) molten salt reactor (MSR), and (6) supercritical-water-cooled reactor (SCWR).[13] The structural materials in these nuclear reactors are challenged by significant amounts of neutron displacement damage at a wide range of temperatures. The operating temperature and neutron exposure of reactors are represented in Figure 2b. The operating conditions for Gen-II reactors are in the range of 550 °C to 600 °C with neutron doses lower than 50 dpa, whereas Gen-III reactors can bear neutron doses up to 75 dpa.[14] In comparison to Gen-II and Gen-III nuclear reactors, Gen-IV reactors have higher operating temperature ranges of up to 800 °C with high neutron doses of up to 200 dpa. High burnup of fuel in Gen-IV reactors may only be achieved if their structural materials and cladding, duct, and wrapper materials are working satisfactorily under exposure to very high temperatures and neutron doses.

### 1.3. Materials selection for nuclear fuel cladding

Gen-IV nuclear reactors are a set of theoretical designs that are currently being developed to utilize advanced fuels and that could operate at high burn-up fuel rates. These reactors are being designed in such a way that reduces nuclear waste and costs while improving energy efficiency, sustainability, and safety.[4] Multiple constraints are being considered for the nuclear reactor's core design, such as operating temperature and thermal efficiency with maximum neutron absorption. During the fission of U atoms, radioactive materials are produced, which emit neutrons and $\gamma$ rays along with $\alpha$ and $\beta$ particles. These particles are hazardous to human beings. The fuel-clad tube separates radioactive materials (fuel rods) from the coolant, which also keeps fuel rods cool. The clad tube is an integral part of the core structure materials of the nuclear power plant. The typical clad tube within a nuclear reactor core can be seen in Figure 3.

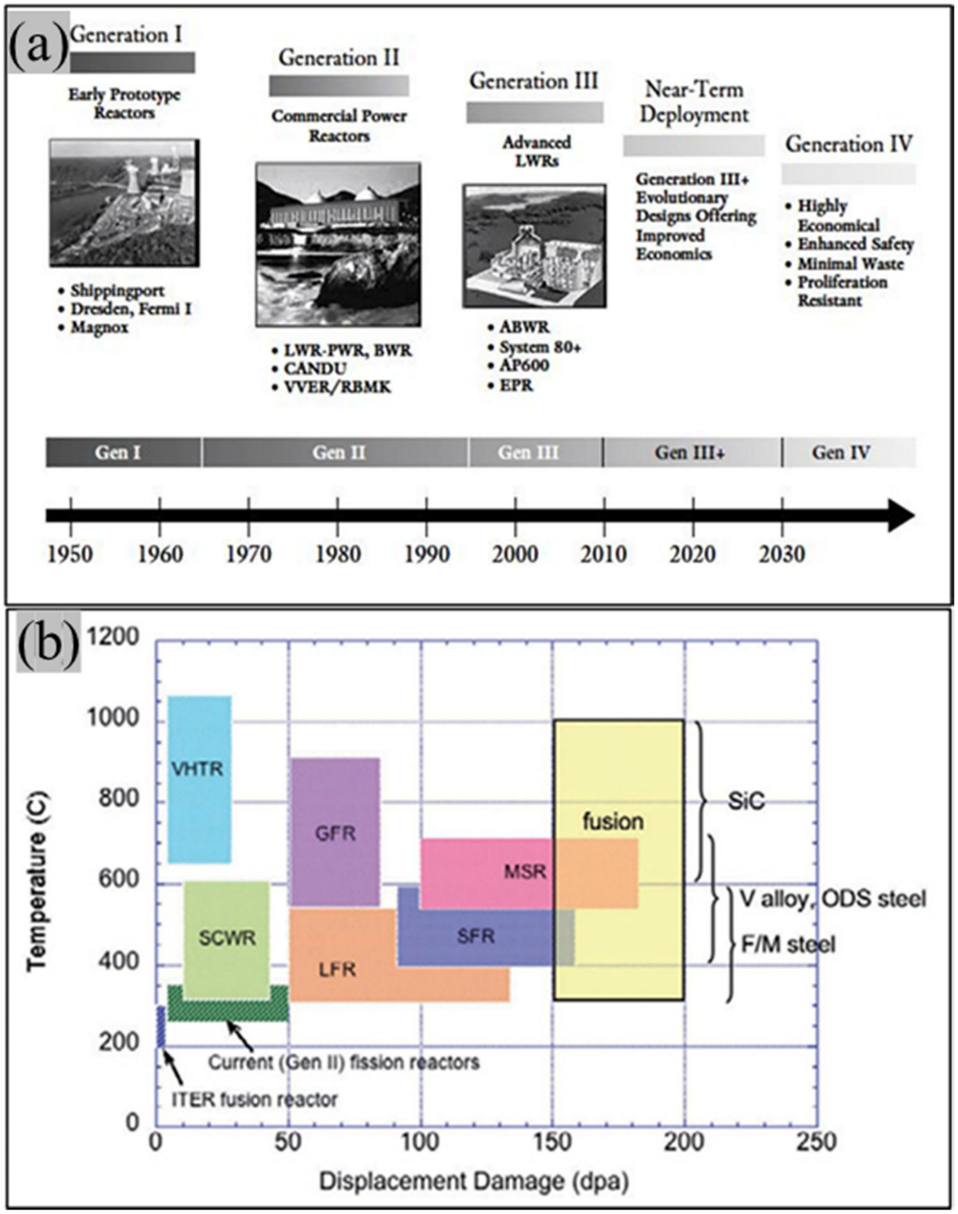


**Figure 2.** (a) Evolution of nuclear fission reactors[10] (Reprinted from U.S. Department of Energy, Office of Nuclear Energy, "Generation IV Nuclear Energy Systems: Program Overview" (Department of Energy, n.d.), http://nuclear.energy.gov/genIV/neGenIV1.html. and (b) Operating temperature with neutron exposure for different reactors.[12]

Because of direct contact with fuel, these tubes are subjected to very harsh service conditions, summarized below:

1. Very high fuel burn rate of approximately 150–250 GWd/tU (burnup in Gen-II and Gen-III reactors is about 60 GWd/tU), due to this clad tube being exposed to very high temperature.[16]
2. Extremely high neutron dose up to 200 dpa.
3. Highly corrosive environment due to direct contact with the coolant (water, sodium, lead).
4. Thermal and irradiation creep due to extremely high temperatures.

Therefore, after deciding on the other aspects of the nuclear reactor's core, such as nuclear fuel, moderator, and coolant materials, the selection of the clad tube materials should be done. Hence, the fuel cladding material selection is constrained by several design limitations. These constraints include that clad tube materials should be transparent to the neutrons, i.e., they should have a low value of neutron cross-section area (such as magnesium (Mg), beryllium (Be), silicon (Si), aluminum (Al), and zirconium (Zr)) which minimizes the neutron losses. Further, these clad materials should have good compatibility with the coolant (such as Pb, Na, and $H_2O$),

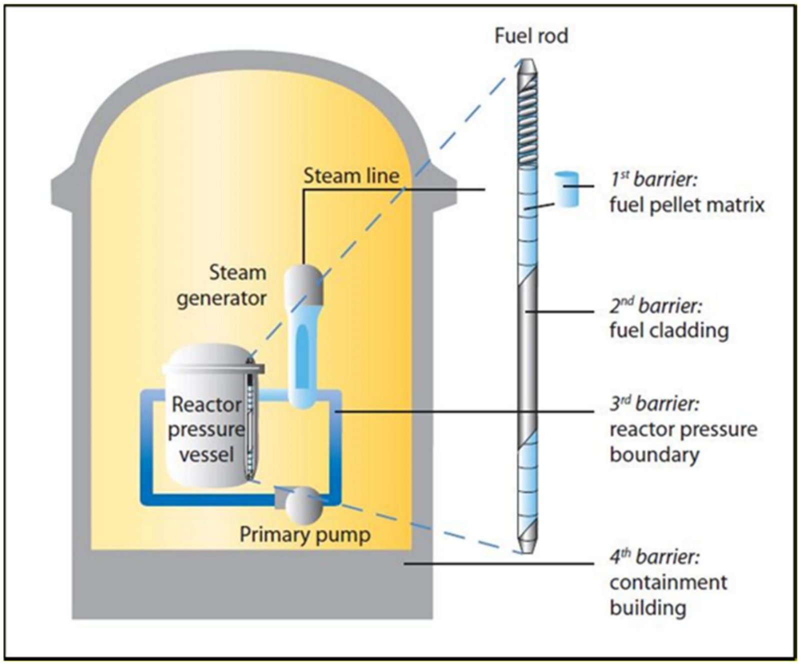


Figure 3. Typical clad tube within the nuclear reactor core.[15]

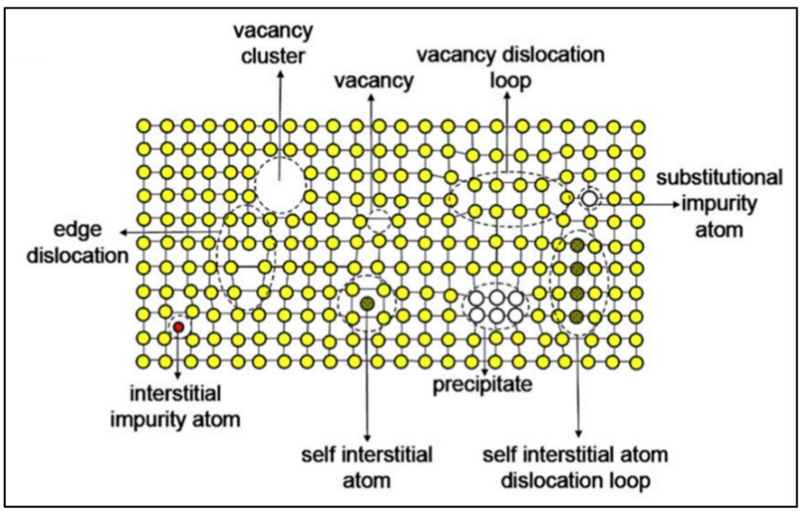


Figure 4. Irradiation induced defects in the crystal lattice.[17]

moderator, and fuel materials.[4] During the fission reaction, the generated neutrons may interact with the crystal lattice of the clad materials in several ways, producing many crystal defects like points, lines, and volumes, such as dislocation loops and voids. The various irradiation-induced defects are shown in Figure 4. This leads to degrading the mechanical properties of the clad materials when the operating temperature is below 40% of the homologous temperature.[17] Exposure to a high dose of radiation may change the phase compositions by the redistribution of alloying elements that can alter phase stability and affect the kinetics of phase transformation. The clad tube materials should withstand high temperatures and high radiation doses, which will help to increase the efficiency of the nuclear power reactors.[4] At a high service temperature, it should have higher creep strength.[18] The thermal conductivity of these materials should be high, whereas their coefficient of thermal expansion should be low to minimize the thermal stress between clad tubes and fuel pellets.[4,19] Finally, clad tube materials should have good corrosion, oxidation, and swelling resistance to the exposed environment.[4]

Hence, the selection of clad tube materials for Gen-IV nuclear power depends on the following factors, which need to be kept in mind:

1. Dimensional stability against high-temperature irradiation.
2. Swelling resistance against exposure to extremely high temperatures.
3. High-temperature mechanical properties (such as tensile strength, ductility, creep strength, fracture toughness, and fatigue).
4. Compatibility with the coolant and moderator.
5. Thermal conductivity and coefficient of thermal expansion.
6. Corrosion and oxidation resistance (with the surrounding environment).
7. Radiation damage.
8. Feasible manufacturing route.

All these factors have their own significance; for instance, in some applications, low thermal conductivity may decrease reactor efficiency, whereas void swelling at high temperature causes dimensional changes, which makes loading and unloading problematic.

High operating temperature and intense neutron radiation in severe environmental conditions are major obstacles to the selection of materials for Gen-IV nuclear reactors. The alteration of microstructure by fast neutron irradiation may affect both the creep and corrosion resistance of the cladding materials during service. This creates more obstacles to a full assessment of the in-service properties of the newly developed materials. Hence, a comprehensive assessment of these newly developed materials for advanced nuclear reactors in a prototypic working environment would be a key step toward their commercial acceptability.[12] Several materials have already been proposed as candidate fuel cladding materials for Gen-IV reactors, which include zirconium alloys (Zr alloys), nickel-based super alloys (Inconel and Incoloy), refractory metals, ceramic materials (SiC and ZrC), austenitic stainless steels (ASS), ferritic–martensitic stainless steel (F/M steel), and oxide dispersion strengthened (ODS) alloys, as shown in Table 1.[4] Most of the materials need further development to increase the nuclear reactor's operating life. Zr alloys were initially not considered for SCWR reactors because of their inadequate mechanical strength and corrosion resistance at high service temperatures and radiation levels. Corrosion of Zr alloys typically produces a multi-layered oxide structure due to a repeated process of oxide growth followed by a transition to another layer. Hence, understanding and controlling its passivation are necessary. Apart from this, these cladding materials limit service temperature up to 400 °C.[20–22] New Zr alloys for nuclear cladding should overcome not only corrosion resistance at higher temperatures but also increase creep and radiation resistance requirements. Ni-based superalloys, particularly Inconel 690, 625, and 718, offer good creep rupture life and high-temperature strength. These are being considered as cladding materials for SCWR. The problem associated with Ni-based alloys is that they have low radiation resistance due to their FCC matrix. Hence, these alloys exhibit radiation embrittlement, swelling, and phase instability on prolonged exposure to high temperatures.[22] During neutron irradiation (at around 15 dpa) at temperatures between 400 and 600 °C, these alloys face significant swelling and a substantial reduction in ductility. Radiation-enhanced intermetallic precipitation and growth of carbides such as $M_{23}C_6$ and $M_6C$, along with the formation of He bubbles at the grain boundaries, were linked to ductility loss.[4] Refractory alloys are considered as cladding materials for LFR and GFR reactors with a maximum operating temperature of 850 °C. These alloys have a melting temperature above 1850 °C, but due to their high values of neutron absorption cross-section area, most of them are not considered cladding materials. They also exhibit poor oxidation resistance and low-temperature radiation embrittlement and are difficult to fabricate.[22] However, refractory alloys are being considered for NASA's nuclear reactor, which can operate at a temperature above 800 °C.[23] Ceramic materials (SiC and ZrC) are being studied as promising cladding materials for LFR, VHTR, and GFR reactors that will be subjected to high temperatures (above 850 °C) and fast neutron radiation.[24] The fuel cladding material for the GFR reactor faces two major challenges: strong resistance to fast neutron damage and high-temperature exposure (up to 1600 °C). Chemical compatibility with the lead (Pb) or lead-bismuth (Pb–Bi) coolant and the mixed nitride fuel is the main concern for the LFR reactor. SiC is considered the candidate material for GFR and LFR reactors.[4] ZrC and SiC have been proposed as coating components for the TRISO fuel in the VHTR reactor to offer mechanical stability and serve as a major diffusion barrier to the emission of fission products.[25] Under some conditions, SiC is prone to oxidation, and its resistance to stress corrosion cracking (SCC) should be examined further. In order to examine the utilization of SiC as an in-core structural material of

**Table 1.** Showing reactor type its fuel, coolant, moderator with operating temperatures and suitable cladding materials.[4]

| Reactor type | Fuel | Coolant | Moderator | Neutron spectrum | Core outlet temperature (°C) | Dose (dpa) | Candidate cladding materials |
|---|---|---|---|---|---|---|---|
| Super critical water-cooled reactor (SCWR) | $UO_2$ (thermal) MOX (fast) | Water | Water | Thermal or Fast | ~500 | 10–40 | Zr alloys<br>Austenitic stainless steels (ASS 304,310, 316, and 316 L)<br>F/M steels (9Cr–1Mo)<br>ODS alloys (F/M (9–12 Cr) ODS steels -Not suitable in highly corrosive environment, Ferritic (12–18Cr) ODS steels- Suitable in highly corrosive environment)<br>Ni based superalloy (IN617, IN625 and IN718) |
| Sodium cooled fast reactor (SFR) | UPuC/SiC U-Pu-Zr MOX | Liquid Na | – | Fast | ~500 | 90–160 | F/M steels<br>ODS alloys |
| Lead- cooled reactor (LFR) | Nitrides, MOX | Liquid Pb alloys | – | Fast | 550–800 | 50–130 | Austenitic stainless steels<br>F/M steels<br>ODS alloys<br>Sic<br>Refractory alloys |
| Gas-cooled fast reactor (GFR) | (U, Pu) O2 Carbide fuel (U, Pu) | He | – | Fast | ~850 | 50–90 | ODS alloys<br>Refractory alloys<br>SiC |
| Molten salt reactor (MSR) | Salt | Molten salt | Graphite | Thermal | 700–800 | 100–180 | No cladding |
| Very high temperature reactor (VHTR) | TRISO UOC | He | Graphite (thermal) | Thermal or Fast | 1000 | 7–30 | ZrC coating<br>SiC coating |

Gen-IV reactors, significant challenges require additional R&D operations.[12] The main disadvantage of ZrC over SiC is that it has lower mechanical strength as well as oxidation resistance. ZrC has a higher thermal expansion coefficient (30% higher than SiC), and a complex fabrication route limits its use.[25,26]

Austenitic stainless steels (ASS) are being considered as promising materials for fuel clad tubes of SWRC and LFR reactors up to the operating temperature of 600 °C. These steels exhibit good creep strength and corrosion/oxidation resistance at elevated temperatures.[27] However, these steels are prone to radiation damage such as extensive void swelling, radiation-induced segregation, He embrittlement, irradiation creep, and microstructural instability even at low neutron irradiation doses.[22] Depletion of Cr at the grain boundaries due to irradiation lowers its corrosion resistance to water-cooled reactors. Further, austenitic steels (FCC) have lower void swelling resistance than ferritic (BCC) and martensitic (BCT) steels.[4] The void swelling for these steels is mainly dependent on irradiation temperature, neutron flux, and applied stresses.[18] The swelling has been reduced significantly by extending the incubation period, adding stabilizing materials, altering the chemical composition, and performing cold work, as illustrated in Figure 5.

These major challenges for austenitic stainless steel (ASS) are still present; therefore, other families of stainless steels are being considered that seem to likely to meet the design constraints for the selection of clad tube materials for Gen-IV reactors.[12,25,28] Ferritic/martensitic (F/M) stainless steels having 9 to 12% Cr are feasible clad tube materials for nuclear reactors such as SFR, SCWR, and LFR with operating temperatures up to 500 °C.[29,30] In comparison to austenitic stainless steels, F/M stainless steels have better thermal properties, such as higher thermal conductivity and lower thermal expansion. These steels have good compatibility with heavy liquid metal coolants such as Pb and have good swelling resistance and He-embrittlement resistance even at high doses of displacement, up to 150 displacements per atom (dpa). Due to their BCC/BCT crystal structure and tempered martensitic microstructure, these steels provide a high density of irradiation defect sinks.[22,24,29] For the fast reactor program, the United States has chosen HT-9 steel, whereas similar types of steel were adopted by Europe and Japan, namely EM12, DIN 1.4914, and PNC-FMS (ferritic-martensitic steel) steels, respectively.[18] The developed conventional 9Cr–1Mo F/M steels and their variants are suitable for applications below 500 °C in liquid metal-cooled fast reactors (LMCR). However, with the advancement of fusion

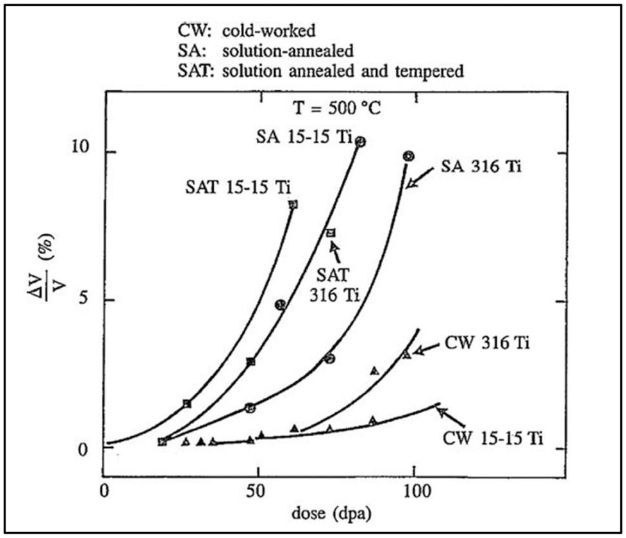


**Figure 5.** Improvement in swelling of austenitic stainless steel by application of various processes.[18]

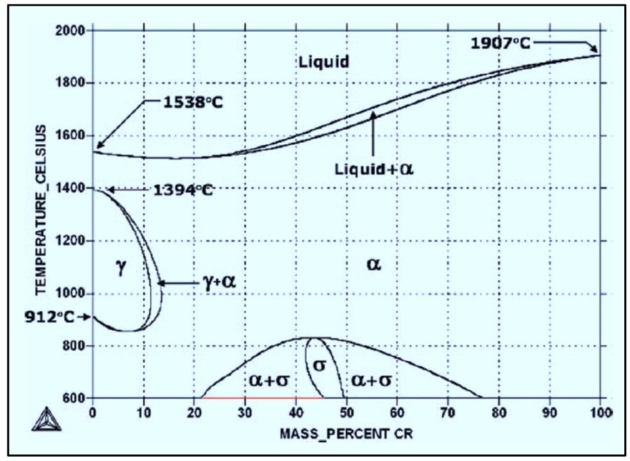


**Figure 6.** Fe-Cr phase diagram.[40]

power, there is a need for new steel grades that can withstand high-energy neutron irradiation and reduce hazards from induced radioactivity. To address this, extensive global research has led to the development of Reduced Activation Ferritic-Martensitic (RAFM) steels for fusion reactor application. Extensive research has also been conducted on RAFM steel in the past.[31,32] Additionally, in Europe, in the late 90's, specifications to produce a European RAFM steel were stated for EUROFER97,[33] which is considered a viable structural material to be tested during the operation of the International Thermonuclear Experimental Reactor (ITER).[34,35]

In reality, both austenitic and ferritic–martensitic steels appear to have serious limitations in the extremely demanding design requirements for commercial fast reactors, the former due to void swelling and the latter due to insufficient creep strength. Introducing a high number density of stable nano-sized oxide particles that pin dislocation motion is one way to improve high-temperature strength in F/M steels. Significant progress has been made in the development of oxide dispersion strengthened (ODS) F/M steels and ferritic alloys,[12,22,36–38] as detailed in the next section.

## 2. Development of ODS steels

Within the European framework of GETMAT (Gen-IV and transmutation materials), emphasis has been placed on developing Fe–Cr oxide dispersion strengthened (ODS) steels for structural in-core uses in Gen-IV reactors. ODS steels show good high-temperature (800 °C) properties and can be utilized as high burnup fuels. The high-temperature mechanical properties of these steels are improved by the introduction of nano-sized yttria ($Y_2O_3$) particles into the Fe–Cr matrix through mechanical alloying. Yttria is preferred over nitrides and carbides because of its high thermal stability at elevated temperatures. It also provides better strengthening than other oxides such as $TiO_2$, $MgAl_2O_4$, $ZrO_2$, and $La_2O_3$.[39] Mechanical alloying with a small addition of Ti refines the reinforcing oxides by forming complex oxides ($Y_2Ti_2O_7$ and/or $Y_2TiO_5$). These oxide particles also act as trapping sites for defects such as interstitials and vacancies, which may form during irradiation. There are mainly two categories of ODS steels being developed as candidate materials for SFR, SCWR, LCR, and GCR reactors: ferritic martensitic (F/M) and ferritic oxide dispersion-strengthened steels. F/M ODS steels have Cr content between 9 and 11% (all compositions in weight %), whereas ferritic ODS have 12 to 20% Cr content. Sigma ($\sigma$) phase formation takes place on increasing the Cr content to values greater than 20%. F/M ODS steels are phase transformable, whereas in ferritic ODS steels, the ferrite phase is stable up to melting. The phase diagram of Fe-Cr steels is shown in Figure 6.

### 2.1. Ferritic-martensitic (9–11Cr) ODS steels

F/M ODS steels have received considerable interest as they have shown great potential to be used as the structural material of advanced fusion reactors since they were recognized as possible structural materials in modern sodium-cooled fast reactors. F/M ODS steels are also being developed for fuel cladding in fast reactors as well as for fusion reactor structural materials.[41,42] They are promising materials for high-temperature applications due to the uniform distribution of thermally stable nanoscale oxide particles in the ferritic-martensitic matrix.[43] Because of the highly stable nanosized oxide particles, 9Cr-ODS steels are more resistant to neutron irradiation embrittlement and void swelling than conventional ferritic/martensitic (F/M) steels.[44] Japan Nuclear Cycle (JNC) Development Institute has been doing research and development of F/M ODS steels as a potential cladding material for advanced fast reactors since 1987.[43] The aim of this technological development is to study the mechanical alloying process and the influence of alloying elements on high-temperature mechanical properties. Based on the findings of those investigations, hot extrusion (HE) and warm rolling (WR) methods were used to manufacture thin-walled cladding in 1990. This preliminary investigation found that the developed clad tube had significantly lower creep rupture strength in the biaxial hoop direction than in the longitudinal direction due to the anisotropy produced during manufacturing. These undesired mechanical properties are due to grain boundaries sliding among the extremely elongated grains parallel to the rolling direction. To make equiaxed grains in 9Cr ODS steels, various heat treatment processes need to be employed. Arup et al.[45] conducted a microtexture analysis on sections of consolidated F/M ODS steel rods, revealing a [110] fiber texture characteristic of bcc metal rods. In contrast, the ODS steel tube exhibited randomly oriented ferrite grains. The absence of texture in the tube is beneficial for the intended application. These steels are used below the temperature of 800 °C to avoid high-temperature phase transformation to austenite. The low-Cr ODS steels do not meet the requirements of the fast breeder nuclear reactors due to low corrosion and oxidation resistance.[46,47] Hence, high-Cr steels are being developed for advanced nuclear reactors.

### 2.2. Ferritic (12–20Cr) ODS steels

Ferritic ODS steels are more suited for fast breeder nuclear reactors because of their high corrosion and oxidation resistance and their high irradiation resistance. Furthermore, they also have good compatibility with the sodium coolant and are the most suitable clad tube materials for these advanced nuclear reactors. These steels do not undergo phase transformation up to melting (Figure 6). Most of the work on ferritic ODS steels is done between 12 and 16% Cr content.[48–51] This is because there is an increase in anisotropy in mechanical properties with increasing Cr content.[37] Also, the sigma phase may form at Cr contents of 20% and above. Reprocessing unspent fuel is one of the main requirements of the nuclear program. The dissolution of clad tubes in nitric acid during the reprocessing of spent fuel may impose serious difficulties on the filtration of unspent fuel.[46] Ferritic steels having low chromium (<14%) content may not meet the requirements of fast breeder reactors (FBRs). Hence, 18Cr ferritic ODS steels, which are more resistant to acid attack, are being developed for advanced nuclear reactors.[47] Limited literature is available on 18Cr ferritic ODS steels.[52,53] Besides excellent swelling resistance, these steels also exhibit very good strength and ductility at room temperature and creep strength at temperatures of 650 °C and above. This is due to the presence of homogeneously distributed Y-Ti-O oxide particles in the ferrite matrix. In comparison to martensitic ODS steels, these steels are designed to withstand a wider range of temperatures up to 1100 °C.[54] In the 1980s, the International Nickel Company (INCO) manufactured extruded ODS ferritic steels named MA956 and MA957. The chemical compositions of these two ferritic ODS steels are Fe-20Cr-4.5Al-0.33Ti-0.5$Y_2O_3$ and Fe-14Cr-0.3Mo-1Ti-0.25$Y_2O_3$, respectively. The alloy MA957 has similar compositions to alloy MA956 but has lower Cr content and the absence of aluminum. High-chromium ODS steel retains its integrity during spent fuel refining and is a viable choice for Gen-IV nuclear reactors.

### 2.3. Austenitic ODS steels and nickel-based ODS alloys

Austenitic stainless steel reinforced with oxide particles can also be a viable option for nuclear power reactors, offering excellent corrosion, oxidation, and creep resistance. However, currently the void swelling resistance of the austenitic ODS steels is comparatively lower than the ferritic ODS steels. These are under consideration as a promising material for Gen-IV nuclear power reactors. Both ferritic and austenitic ODS steels are available in various grades of austenitic stainless steels, such as 304, 310, 316, and 316 L, with varying amounts of $Y_2O_3$ and Ti used as dispersoids. These steels are expected to exhibit superior high-temperature properties and improved resistance to void swelling compared to ferritic ODS steels. For the last twenty years and more, ferritic ODS steels have been widely studied and utilized in the nuclear industry. Extensive research has been conducted on their microstructure, nanocluster composition, irradiation resistance, and high-temperature properties (tensile and creep), providing a detailed understanding of their behavior and performance. However, research on austenitic ODS steels is limited, and further detailed studies are needed before their use in nuclear power reactors.[55]

Nickel-based ODS superalloys, known for their excellent creep strength, are promising candidates for high-temperature applications in gas turbines and Gen-IV nuclear power reactors. Various grades (IN617, IN625, and IN718) of Ni-based ODS super alloys are being developed specifically for nuclear power reactors.[56–58] A conventional non-ODS variant is unstable at high temperatures (greater than 650 °C) due to the coarsening of gamma prime ($\gamma$') and gamma double prime ($\gamma$'') precipitates. At relatively high temperatures (~1000 °C), these precipitates and hard carbides dissolve in the $\gamma$ matrix, which diminishes the precipitation effect in the alloy.[59] This leads to deterioration in the high-temperature mechanical properties of these alloys. To retain the high-temperature strength, nano oxide particles (mainly $Y_2O_3$) are dispersed in the $\gamma$ matrix to restrict the dislocation motion and grain boundary sliding.[60] Over the years, several commercially available Ni-based ODS superalloys have been developed for high-temperature applications, including MA6000, MA754, PM1000, and PM3030. In addition, the literature reports various other Ni-based ODS alloys, such as Ni-20Cr-1.2$Y_2O_3$ wt%,[61] Alloy 617–0.6 wt% $Y_2O_3$,[56] IN-718 with 0.25–1 wt% $Y_2O_3$,[58,62,63] and Hastelloy X ODS alloy.[64] Few nickel-based ODS steels with the addition of iron have also been developed.[58,65]

## 3. Fabrication routes of ODS steels

The oxide dispersion-strengthened steels are generally produced through powder metallurgy (P/M) routes. Due to a significant difference in the melting point and buoyancy issues in the melt, it is difficult to prepare ODS steels through conventional melting processing technique.[66] Mechanical alloying (MA) is a feasible processing technique that provides uniform fine dispersion of oxide particles in the ODS alloys. Currently, MA is the most effective method to disperse yttria into the iron matrix.[67] A schematic fabrication route is represented in Figure 7. Powder processing using mechanical alloying is the preferred route for producing these steels. The powder metallurgy route is used over casting to produce ODS steels because the primary strengthening elements (Y&O) have extremely low solubility and wettability in liquid steel, which leads to inhomogeneous mixing/segregation of oxide particles.[69] During mechanical alloying, the powder particles are repeatedly cold-welded, fractured, and rewelded in a high-energy attritor or ball mill.[70,71] Hard and brittle nanosized yttria ($Y_2O_3$) is dispersed into the steel matrix by mechanical alloying. MA allows uniform oxide particle distribution throughout the matrix. The milled powder is then encapsulated in mild steel cans and degassed at elevated temperatures to prevent oxidation in subsequent manufacturing processes. Further, the milled powders are consolidated through various consolidation techniques such as extrusion, hot isostatic pressing, or spark plasma sintering (SPS).[72–74]

In the extrusion process (HE), powders are heated and forced through a die under pressure, producing a dense product, typically in the form of rods or tubes. The application of high pressure and temperature increases the final density. However, a prolonged consolidation time can lead to grain coarsening in the steel and should be avoided. The hot isostatic pressing (HIP) process involves compressing the powder under high pressure and elevated temperature, resulting in a high-density, near-net-shaped product. This process is time-consuming and may lead to microstructural coarsening. On the other hand, the spark plasma sintering (SPS) method uses high-temperature pressure pulses to rapidly consolidate the powders, producing a dense product with a fine-grained microstructure.[75] However, the product obtained with this route is limited in size. Dash et al.[76] systematically studied the densification kinetics of 18Cr-ODS ferritic steel powder consolidated using SPS. They found that the consolidation temperature at a given stress has a strong effect on densification up to 1323 K (1050 °C), after which it saturates within a narrow range of about 50 K. On the other hand, soaking time had little impact on the final density.

During mechanical alloying, Ti facilitates the dissociation of $Y_2O_3$ particles into Y and O.[77,78] However, some studies suggest that pure $Y_2O_3$ may also undergo structural transformation during mechanical alloying.[79] Upon consolidation of the powders, the dissolved oxygen may react with Y and Ti to form complex Y-Ti-O oxides, such as $Y_2TiO_5$ and/or $Y_2Ti_2O_7$.[78,80] Several researchers have utilized high energy ball milling (HEBM) to synthesize $Y_2Ti_2O_7$ directly from $Y_2O_3$ and $TiO_2$ powders, followed by thermal annealing.[81,82] Gadipelly et al.[83] prepared $Y_2Ti_2O_7$ nanopowder using a microwave hydrothermal method. The resulting $Y_2Ti_2O_7$ can be directly dispersed into the ODS steel matrix during milling.

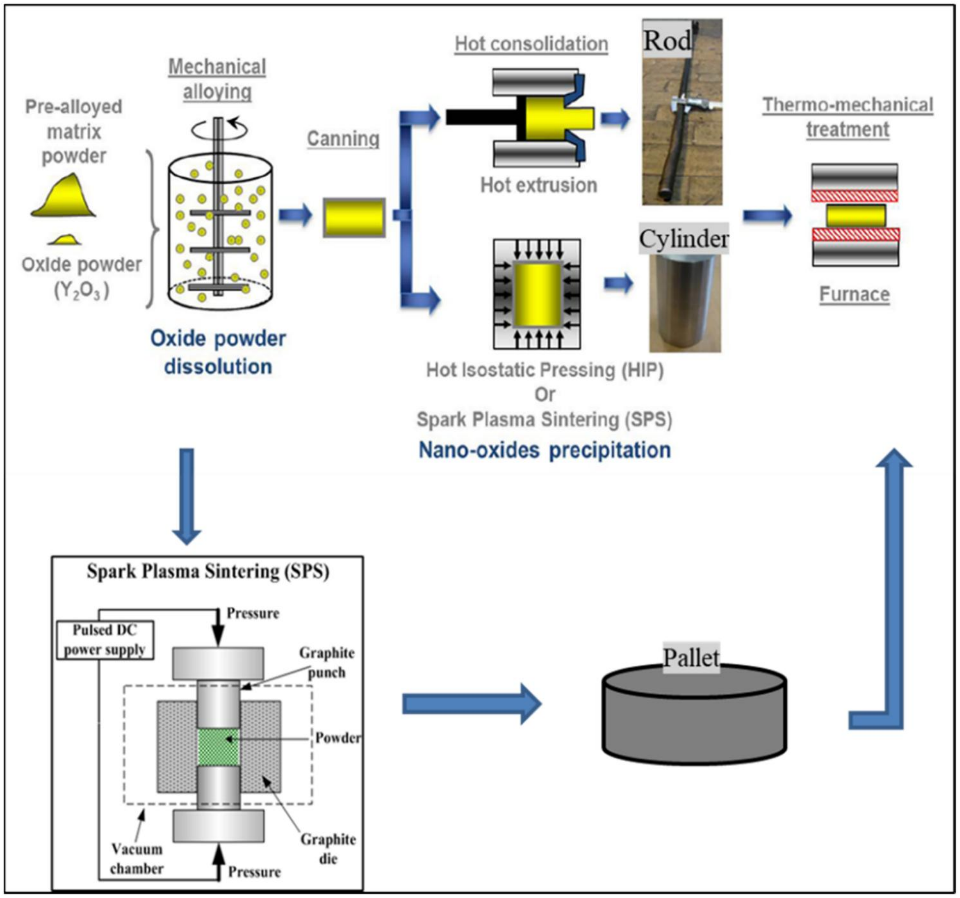


Figure 7. A schematic diagram to produce ODS steel (Adapted from [68]).

The relative proportions of Ti and $Y_2O_3$ play a crucial role in influencing the shape and size of the oxide particles in the ODS alloys.[84] The formation of complex oxides can be explained by the following reactions.

$$Y_2O_3 + 2TiO_2 \rightarrow Y_2Ti_2O_7$$

$$Y_2O_3 + TiO_2 \rightarrow Y_2TiO_5$$

During consolidation of alloy powders, the reaction between $Y_2O_3$ and $TiO_2$ leads to the formation of a complex oxide of Y-Ti-O ($Y_2Ti_2O_7$ or $Y_2TiO_5$). Transformation of the oxide particle from $Y_2O_3$ to $Y_2Ti_2O_7$ or $Y_2TiO_5$ can be understood from the above two equations. It is concluded that the formation of $Y_2Ti_2O_7$ oxide particles requires a higher Ti content compared to $Y_2TiO_5$ for the same amount of $Y_2O_3$. Moreover, when the Y/Ti ratio is high, the formation of $Y_2TiO_5$ particles is more likely than the formation of cubic $Y_2Ti_2O_7$. This behavior was observed by London et al.[77] in their alloy. The formation of $Y_2TiO_5$ oxide particles is more thermodynamically favorable than that of $Y_2Ti_2O_7$. This is attributed to its lower formation energy (3.771 eV) compared to $Y_2Ti_2O_7$ (3.877 eV).[85] Additionally, the formation of $Y_2TiO_5$ (per mole) requires less oxygen than $Y_2Ti_2O_7$.[85] As a result, the amount of excess oxygen present in ODS steels can also influence the proportion and type of oxide particles formed within the matrix.

It has been observed that most of the researchers have added $\sim$ 0.2% Ti for $\sim$ 0.35% $Y_2O_3$ (i.e., $Y_2O_3$/Ti $\sim$ 1.75) in ODS steels to get an effective number of fine complex oxides of Y-Ti-O.[86,87] Various amounts of $Y_2O_3$ and Ti, in ratios such as 1:1[5,41,54,84] and 7:4,[32] have been reported in the literature. Similarly, Parida et al.[88] have used $Y_2O_3$ and Ti in the ratio of 3:1 to study the evolution of various oxides on subsequent annealing. They found that most of the oxide particles after annealing were either $Y_2Ti_2O_7$ or $YTiO_3$. However, there is still no clear research direction on how specific $Y_2O_3$/Ti ratios influence the formation of these oxide phases. An optimized ratio of $Y_2O_3$ and Ti needs to be determined to achieve a fine complex oxide of Y-Ti-O in the alloy matrix.

Extruded materials often exhibit extrusion bands parallel to the extrusion direction.[89] Extruded ferritic ODS steels also show elongated grains, which can lead to anisotropic properties.[90] These steels typically

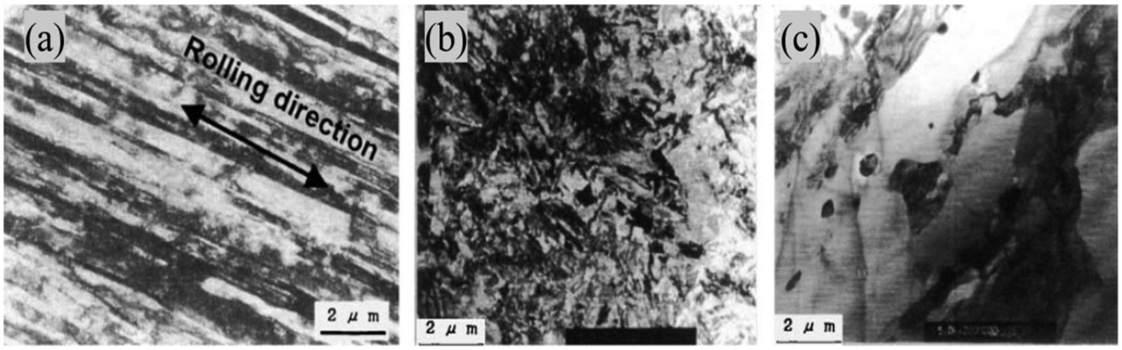


**Figure 8.** Showig TEM image of (a) elongated grain after cold rolling (b) tempered martensitic structure of 9Cr ODS steel and (c) recrystallized microstructure of 12Cr ODS steel.[43]

have a strong preferred crystallographic orientation along the <110> direction in the extrusion axis. This microstructural anisotropy directly affects the mechanical properties of ODS steels. For instance, it is widely recognized that the transverse direction of textured ferritic ODS steels is less ductile compared to the extruded direction. Additionally, the morphology of the grains appears to influence the fracture mode.[91] Various thermomechanical treatments (TMTs) are employed to get desired properties. In this regard, mainly two approaches are considered: In the first approach, F/M ODS steels are heated above the $\alpha$ to $\gamma$ phase transformation temperature and quenched to form tempered martensite, which produces radiation-resistant alloys. In the second approach, an annealing treatment is applied to ODS steels with >12Cr after cold rolling (CR) to develop a recrystallized microstructure.[43] These high-Cr steels do not exhibit the $\alpha$ to $\gamma$ phase transformation, as the ferrite phase is stable up to the melting point. Steels with higher Cr content exhibit better corrosion and oxidation resistance. Examples of microstructures of cold-rolled and heat-treated 9Cr as well as 12Cr ODS steels are given in Figure 8. The final use of the ODS steel is in the form of a clad tube of the nuclear reactors. These clad tubes are manufactured through the pilgering (rolling operation) process. Rolling produces directional properties in the clad tubes. The creep strength in the transverse (hoop) direction is approximately 50% of that in the axial direction.[4] In a nuclear power reactor, the clad tube of the fuel rod is subjected to a biaxial state of stress. Producing ODS steel tubes with equiaxed grains is a challenging task, as the manufacturing process strongly affects the crystallographic texture;[92] then, a recrystallization treatment is attempted to produce equiaxed grains in the ferritic ODS steels.

Ukai et al.[93] have extensively studied the recrystallization of ferritic ODS steels. Recrystallization was used to control the grain morphology of ferritic ODS steels and improve their inferior mechanical performance, as described above. They have suggested that cold rolling would be feasible after the recrystallization treatment, as it softens the materials. Extrusion is the preferred method for the consolidation of ODS steels, in contrast to hot isostatic pressing (HIP), which takes longer. But extruded steels exhibit strong fiber texture <110> in the extruded direction,[94] leading to anisotropy. Even at high temperatures, these ferritic ODS steels do not transform into austenite.[95] Hence, it is impossible to remove texture and alter their grain morphology by quenching and tempering, as is the case for 9Cr ferritic martensitic ODS steels. Therefore, only recrystallization treatments can be applied on 12–18%Cr ferritic ODS steels to reduce anisotropy. It is observed that the recrystallization temperature of ferritic steels is of the order of $0.6\,T_m$ ($T_m$–melting temperature), whereas the recrystallization temperature of ferritic ODS steels is of the order of $0.9\,T_m$.[51,96] Such a high recrystallization temperature is attributed to pinning of grain boundaries by the nanosized complex oxide (Y-Ti-O) particles in ODS steels. Heating to such high temperatures causes coarsening of the oxide dispersoids, which deteriorates the mechanical properties of the alloys.[51] Achieving a fully recrystallized microstructure in ferritic ODS steels with a high-volume fraction of complex oxides is very challenging. They employed extrusion and cold rolling (60% cold rolled) to study recrystallization in several ferritic ODS alloys. They have observed that the amount of yttria and excess oxygen (total oxygen content in steel minus oxygen content in $Y_2O_3$) in ferritic ODS alloys impacts the possibility of recrystallization. They have also observed that ODS steels containing more than 0.27% (mass) yttria are not able to recrystallize even after 60% cold rolling and at a high annealing temperature of 1200 °C (Figure 9).

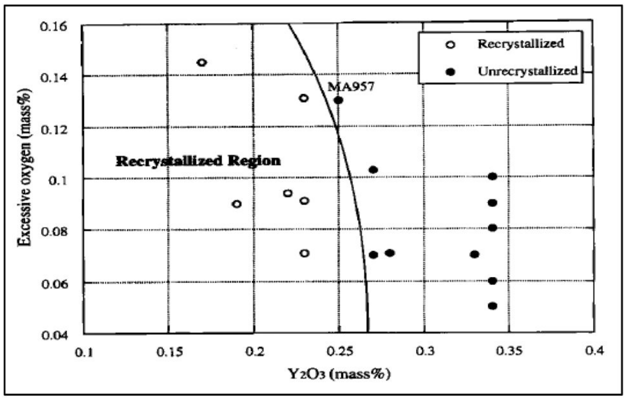


**Figure 9.** Effect of yttria and excess oxygen content on recrystallization of ferritic ODS steel (60% cold worked annealed at 1473K).[93]

Therefore, they have suggested that the maximum yttria content to be added is ~0.25% to achieve improvements in anisotropy of ferritic ODS steels through recrystallization treatments.

F/M ODS steels having higher yttria content up to 1% were investigated by Ukai et al.[97] These steels contained martensitic structure (lath type) with high dislocation densities. The possibility of phase transformation leads to producing high-yttria F/M ODS steels. From the year 1995 to 2000, research was conducted to develop thin-walled ODS cladding.[43] The aim was to prevent crack initiation during the intermediate manufacturing process of clad tubes while ensuring good internal creep strength and ductility with homogeneous grain morphology, using recrystallization and phase transformation as a final heat treatment.[36,98] Recrystallization in extruded ferritic ODS steel followed by rolling was reported by Kim et al.[99] They observed recrystallization and grain growth for the ferritic ODS steel having compositions Fe-12Cr-0.25$Y_2O_3$ and Fe-12Cr-3W-0.25$Y_2O_3$ when annealed at 1250 °C. However, for the ODS steel containing Ti (Fe-12Cr-3W-0.4Ti-0.25$Y_2O_3$) partially recrystallized microstructure was observed. This is due to the formation of complex oxide (Y-Ti-O) in the Ti-containing ODS steels. Recent investigations on ferritic ODS steel revealed that even at high annealing temperatures of 1200 °C, complete recrystallization does not occur.[100–102] Hence, it is very difficult to achieve isotropic properties in ferritic ODS steels with higher (>0.35 wt. %) yttria content through recrystallization treatment. This could be one of the reasons for limited studies being done on high-yttria ferritic ODS steels compared to the F/M ODS steels. Researchers mostly employed homogenization/annealing on the ferritic ODS steel containing 0.35% yttria;

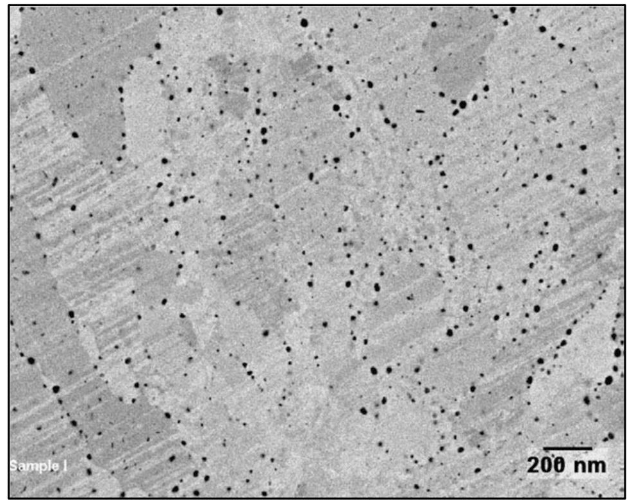


**Figure 10.** Ferritic ODS steel (Fe-14Cr-2W-0.3Ti-0.3$Y_2O_3$) consolidated through HIP showing porosity.[107]

however, recrystallization could be observed only for ferritic ODS steel containing 0.25% yttria.

## 4. Limitations of conventional fabrication routes

The milled ODS steel powders can be consolidated using methods such as hot isostatic pressing (HIP),[103,104] hot extrusion (HE)[39,105] or spark plasma sintering (SPS).[74,106] However, these conventional consolidation processes have some limitations. As discussed in Section 3, a major drawback of the HIP process is the presence of prior particle boundaries (PPBs) in the alloys (Figure 10).[105] As a result, HIPed ODS steels exhibit lower tensile strength and fracture toughness compared to extruded ODS alloys.[105,108] Mechanically alloyed powders may contain porosity inside the powder particles after the milling process.[109] It has been observed that the compressibility of porous powders is less than that of the fully dense powder particles.[109] Hence, consolidation of these powders through only HIPing is difficult.[108,110] Also, the HIPing process is more sensitive to the oxygen content of the powder, as it may lead to the formation of an oxide layer at the surface of the powder particles. This will limit the diffusion process and will lead to residual porosity in the consolidated alloys.[105,108,110] These pores act as stress concentration sites and induce microcracks, which can lead to early fracture, resulting in poor mechanical properties. To achieve homogeneous microstructure and almost fully dense product, thermomechanical treatments (TMTs) such as rolling, forging, and high-speed hydrostatic extrusion (HSHE) have been applied to the HIPed alloys. This improved the interparticle

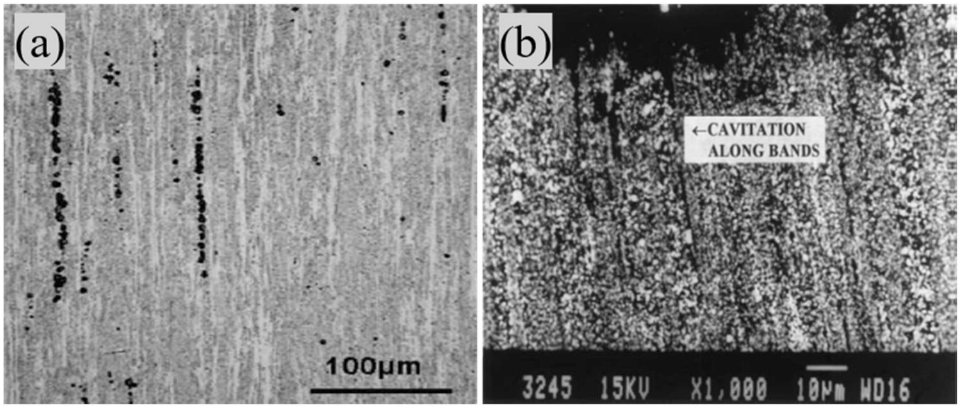


**Figure 11.** (a) Optical microstructure of MA957 ODS steel showing extrusion band along extrusion direction.[113] (b) Cavitation along the extrusion bands in extruded Al-Fe-Ce alloy.[114]

bonding and enhanced mechanical properties of the HIPed alloys.[105,111,112]

Oksiuta et al.[112] have studied the effect of thermomechanical treatments on HIPed ODS steels. They found an improvement in the microstructure and mechanical properties of the ODS steel after TMTs such as hot pressing and hot rolling. The hot-pressing leads to a fine microstructure with equiaxed grains and an improvement in Charpy impact properties. Hot rolling leads to elongated grains and shows improvement in both tensile strength and Charpy impact properties. Consolidation through extrusion requires less time compared to HIP. In extruded alloys, deformation predominantly occurs along the extrusion direction, leading to the formation of stringers and bands (Figure 11a).[89,113] Klueh et al.[113] have reported band formation in extruded ODS alloys, while Prakash et al.[114] have observed similar extrusion bands in aluminum alloys fabricated from pre-alloyed powders (Figure 11b). These bands are undesirable as they can lead to preferential creep cavitation along the bands.[115] Additionally, the presence of bands or stringers can result in microstructural and chemical inhomogeneity in extruded ODS steels, ultimately deteriorating mechanical properties.[113] The extrusion process elongates grains along the extrusion direction, which can induce anisotropy in the material.[89,90] Achieving equiaxed grains in extruded alloys is challenging due to the elongated grains resulting from extrusion. The grain boundaries are effectively pinned by nanosized oxide particles, which hinder the formation of recrystallized equiaxed grains. Further thermomechanical processing is required to achieve equiaxed and recrystallized grains in extruded ODS alloys. Despite extensive thermomechanical treatment, some anisotropy may persist.[97] Moreover, spark plasma sintering (SPS) processes yield products with small sizes, which limits their applicability.

The above-mentioned fabrication routes are used to consolidate ODS steels. Considering their limitations, there is a need to develop alternative fabrication routes to overcome the above-mentioned limitations. The powder forging route provides an alternative fabrication process that may lead to a recrystallized microstructure with nearly isotropic properties. A brief introduction to the powder forging process is given in the subsequent section.

## 5. Powder forging, a new fabrication route

Powder forging is a process in which powders or, sometimes, pre-sintered/sintered powders are hot deformed in a forging press.[116,117] Extensive research (theoretical and experimental) has been carried out to study the mechanisms of deformation and densification in this process.[116] The surface oxides present in the powder particles are broken due to significant shear during the multiaxial material flow. This enhances the bonding between powder particles. Due to multiaxial flow of materials during forging, the properties of the powder-forged alloys are nearly isotropic.[117,118] The density, chemistry, and microstructure of P/M parts influence their mechanical performance. The components produced using traditional P/M (compacting and sintering) routes can be porous. Through powder forging, high energy is imparted to the powders, which densifies the material and leads to ~100% density with high-integrity products.[116,119]

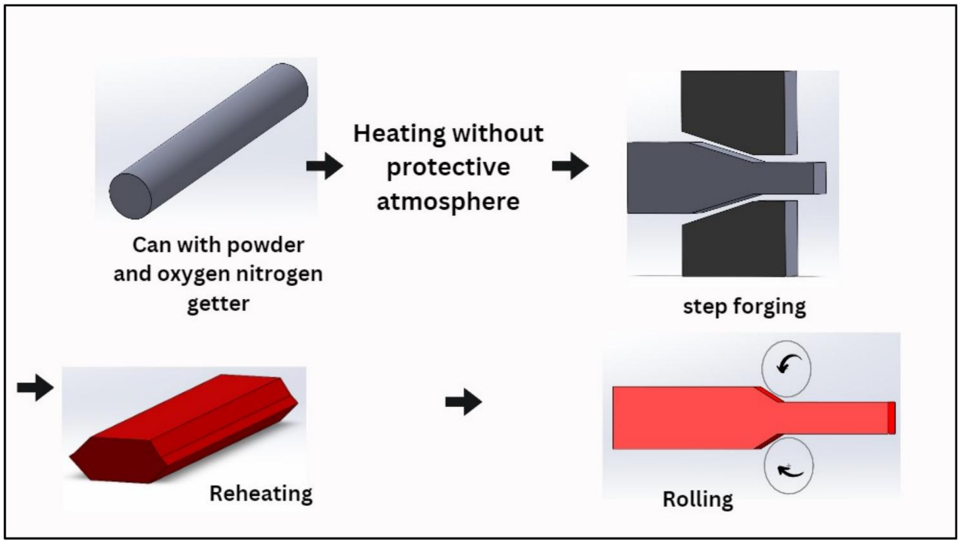


Figure 12. Schematic diagram of powder forging through encapsulation of powders (Adapted from [120]).

### 5.1. Powder forging through encapsulation of powders

Loose powders are filled in a can (generally a mild steel can) and hot consolidated through forging. Olsson et al.[120] have developed the tool steel bar stock using this forging process. They filled powders of desired compositions in a tubular container and heated the sealed container in a furnace. The heated can was directly forged in a forging press. They achieved 99.5% theoretical density in the consolidated material. Further, reduction through rolling increased the density to ~100%.[120] The schematic diagram of powder forging through encapsulation presented by them is shown in Figure 12. They also observed that the density of the forged slab material can be increased by increasing the forging load. However, tool wear (such as die and punch) increases on increasing the applied load.

Bai et al.[121] used direct powder forging of encapsulated powder to consolidate Ni-based superalloys. This process of forging requires lower cost, energy, and time than the conventional pressing and sintering process. The possibilities of formation of PPBs are negligible due to shorter heating time for forging. Further, PPBs network is broken due to material flow during forging. The absence of PPBs may lead to a significant improvement in mechanical properties. Recently, Kumar et al.[117,118] and Singh et al.[122–125] have utilized hot powder forging to fabricate high yttria-containing ODS steels. They have developed a unique route to consolidate the ODS steel powders through direct forging. In this route, the mechanically alloyed powders were filled in specially designed mild steel capsules. The powder-filled capsules of desired compositions are heated in a tubular furnace under a flowing hydrogen gas atmosphere. The heated mild steel capsule was forged in a customized die and punch using a friction screw forging press. The steps taken to consolidate ODS powder through hot powder forging are systematically shown in Figure 13. The clad tube of a fuel rod experiences a biaxial state of stress, making the isotropic properties of ODS steels crucial for advanced nuclear reactors (Gen-IV). The presence of bands or stringers in the clad tube can lead to failure in the transverse direction. These bands are undesirable because they can serve as sites for preferential creep cavitation. To mitigate these issues and improve the longevity of the power plant, consolidation through the hot powder forging route is advantageous. Powder-forged ODS steels exhibit a recrystallized microstructure that enhances isotropic mechanical properties.[118] By employing powder forging instead of extrusion, the anisotropy associated with high Cr content can be reduced. Powder forging avoids the formation of prior particle boundaries (PPBs) and extrusion bands (stringers), leading to improved material properties.[118,119]

## 6. Crystallographic texture control for ODS clad tubes

Clad tubes are generally manufactured by hot extrusion followed by cold forming processes such as pilgering. However, pilgered samples exhibit elongated grains and pronounced texture formation, with a preferential orientation along the rolling direction. Although heat treatment transforms the grain morphology from elongated to nearly equiaxed, the well-

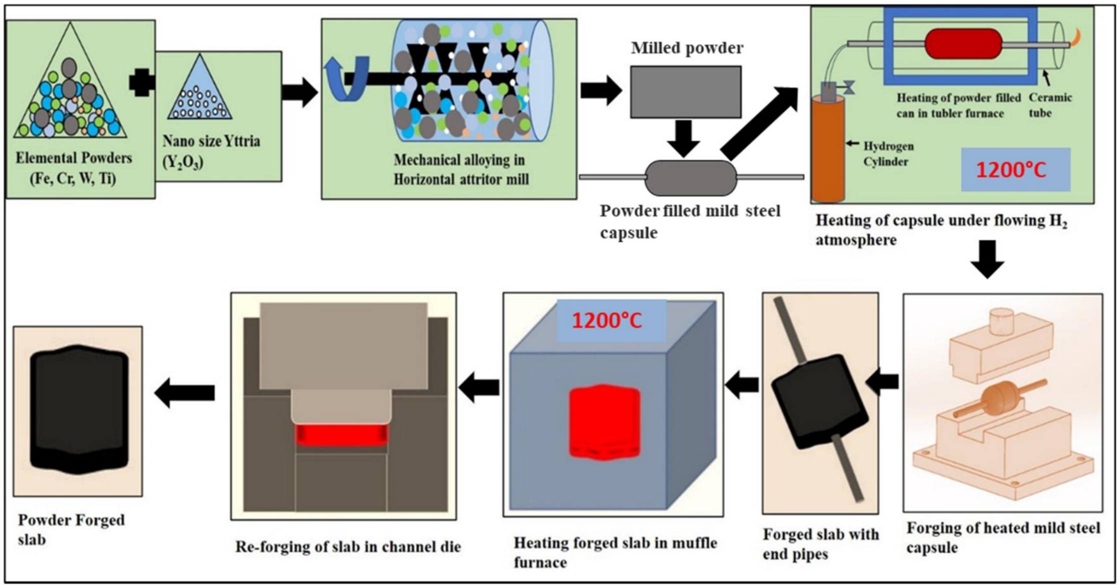


**Figure 13.** Schematic diagram of powder forging through encapsulation of powders (Authors own creation).

known $\alpha$-fiber texture remains present. The exceptional high-temperature stability of the $\alpha$-fiber texture is primarily due to a crystallographic structure memory effect during phase transformations. Vakhitova et al.[126] extensively studied the evolution of crystallographic texture in 9Cr ODS steel during the pilgering process. They investigated two different processing routes: one with and one without intermediate heat treatment between rolling steps. This heat treatment is intended to soften the material and prevent damage during subsequent fabrication steps. Additionally, it helps in reducing mechanical anisotropy, which is a major concern in the production of ODS steels for service applications. Several studies have investigated the ridging phenomenon in ferritic stainless steels, aiming to predict texture evolution and surface profile strain anisotropy using electron backscatter diffraction (EBSD) orientation maps and crystal plasticity models. Dash et al.[127] studied the texture evolution of extruded 18Cr ferritic ODS steels under high-temperature uniaxial compression testing (at 1150 °C and a strain rate of 0.01/s). They employed viscoplastic self-consistent (VPSC) constitutive models to predict variations in texture intensity resulting from the deformation of hot-extruded steel. Initially, an $\alpha$-fiber texture was observed along the extrusion direction (ED) in the microstructure. Furthermore, the VPSC-5 model was used to predict the degree of texture evolution after deformation along both the ED and the transverse direction (TD), relative to the initial condition. The model predictions indicated that deformation along the ED leads to the development of a dominant $\gamma$-fiber texture, in contrast to the TD.

Diminishing or modifying the crystallographic texture in ODS cladding tubes significantly improves mechanical isotropy and enhances the performance of nuclear power plants. Various thermomechanical processing (TMP) methods and heat treatment cycles are applied to the cladding tube materials to achieve this goal. To mitigate texture in ODS cladding tubes, a combination of optimized thermomechanical processing (such as intermediate annealing during pilgering and modified rolling sequences),[128] precise heat treatment cycles (such as recrystallization annealing and stress relief annealing),[129] and alternative fabrication techniques (such as cold spray deposition, friction stir processing, and powder forging).[109,130–132] Apart from this, milled powder particles also exhibit texture, which may need to be minimized before the consolidation process.[133] These techniques show great promise in achieving a more isotropic microstructure, thereby enhancing the performance of ODS cladding tubes in nuclear reactor applications.

## 7. Effect of alloying additions on ODS steels

The primary alloying elements for ODS alloys are carbon (C), chromium (Cr), tungsten (W), titanium (Ti), and yttrium (Y). On the other hand, silicon (Si), phosphorus (P), sulfur (S), and argon (Ar) are considered impurity elements. In the Fe-Cr phase diagram, C is an austenite stabilizer that increases the austenite loop, whereas Cr, W, and Ti are ferrite stabilizers and suppress the austenite loop. C has very small solubility in the ferrite, which results in the formation of carbides. Cr also offers excellent corrosion resistance. At least 11% Cr (by weight) is required to generate a protective passive layer of chromium oxide on the alloy's surface. On increasing the Cr content, the oxidation as well as corrosion resistance of steel increases significantly.[134] ODS steels containing >11%Cr are totally ferritic (Figure 6). These high-Cr ODS steels are preferred because of their high corrosion and oxidation resistance.[23,54] However, if the added Cr content is above 19%, then there may be the formation of sigma phase (Figure 6), which is brittle in nature and may deteriorate the mechanical properties. Hence, to improve corrosion and oxidation resistance, most of the researchers have used a maximum of 18%Cr in the ODS steels.[53,86,117,118,135] Moreover, Cr concentration also affects the mechanical properties. Oksiuta and Baluc[136] have found that the steel containing 14%Cr (Fe-14Cr-2W-0.3Ti-0.3$Y_2O_3$) exhibited higher tensile strength but lower ductility than 12%Cr (Fe-12Cr-2W-0.3Ti-0.3$Y_2O_3$) ODS steels. Li et al.[137] investigated the effect of Cr in the Fe-xCr-1W-0.5Ti-0.35$Y_2O_3$ ($x = 12$, 16, and 18%) alloy consolidated through HIP followed by thermomechanical treatments. They observed that the maximum strength occurred at 16Cr content, above which the strength decreased. They also found that in high-Cr alloys, a Cr-enriched phase forms, which reduces the ductility of ODS ferritic steels. These Cr-rich phases are hard and brittle in nature, which may decrease the homogeneity, ductility, and density of ferritic ODS steels. Cr can lower the fracture toughness by forming a $\delta$-ferrite phase in 12%Cr-ODS steel. C and Mn may be added to the steel to suppress the formation of $\delta$-ferrite. This leads to extensive precipitation of $M_{23}C_6$, which may reduce the fracture toughness. Along with this, the chi ($\chi$) phase might also form during irradiation, which promotes embrittlement in the steels.

W is added to ODS steels to provide solid solution strengthening.[99] It also acts as a ferrite (BCC) stabilizer and improves high-temperature strength. The maximum 2% W is added to avoid the Laves phase ($Fe_2W$) formation in the steel matrix. This intermetallic phase is highly brittle in nature, which causes embrittlement in the steel. The formation of Laves phase reduces W in the solid solution, which may decrease high-temperature strength of the materials. Narita et al.[138] added 2% W without any Laves phase precipitation in 9%Cr ODS steels and found significant improvement in the tensile strength at high temperature. The variation of tensile strength with the addition of W is represented in Figure 14.

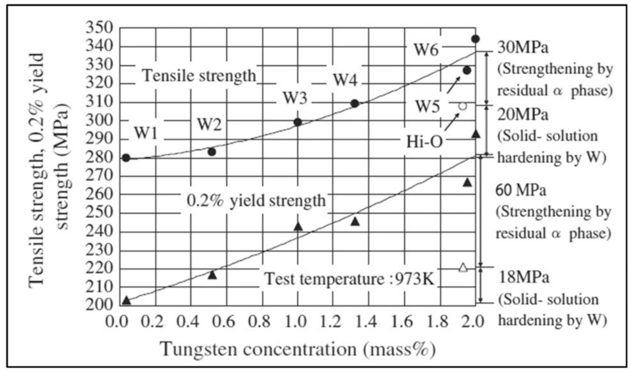


**Figure 14.** Variation of tensile and yield strength with tungsten addition in 9Cr ODS steels.[138]

The addition of Ti significantly increases the tensile and creep strength of the ODS steel. Ti is well known for refining the dispersoids in the ODS steels.[93] It also helps dissolution of the yttria ($Y_2O_3$) during mechanical alloying and forms complex (Y-Ti-O) nanosized oxides during the consolidation process. These nanosized oxides effectively pinn dislocations, which enhance the tensile as well as creep strength of the alloys.[41] The effect of other alloying elements such as V, Nb, and Zr was also extensively studied in 12Cr ODS steel. It is observed that in the case of the alloy not containing Ti, the dispersoid sizes were found to be around 10 nm as compared to the initial size of 20 nm $Y_2O_3$ particles added for alloying.[43] A small amount of Ti addition leads to a significant reduction in the dispersion size down to 3 nm.[43] The reduction in dispersoid size significantly improves the creep strength of the 12Cr ODS steel. Other refining elements such as V, Nb, and Zr are found to be less effective than Ti for refining the dispersoids. The effect of these elements on refining the dispersoids in a 12Cr ODS steel is represented in Figure 15a. Dark field TEM images of the oxide particles and corresponding size distribution with different yttria contents of F/M ODS steels are also shown in Figure 15b-d.

Kim et al.[99] have studied the effect of Ti in 12Cr ODS steels. They observed that the Ti-containing alloy 12YWT (Fe-12Cr-3W-0.4Ti-0.25$Y_2O_3$) exhibited higher yield strength at elevated temperatures than the alloys 12Y1 (Fe-12Cr-0.25$Y_2O_3$) and 12YW (Fe-12Cr-3W-0.25$Y_2O_3$), which do not contain Ti (Figure 16). This is due to the formation of a fine dispersion of complex oxides of Y-Ti-O. The large number of these complex oxides considerably reduces their interparticle distance. The movement of dislocations is effectively restricted by this dispersion of fine oxide particles, thus improving the strength. Oksiuta and

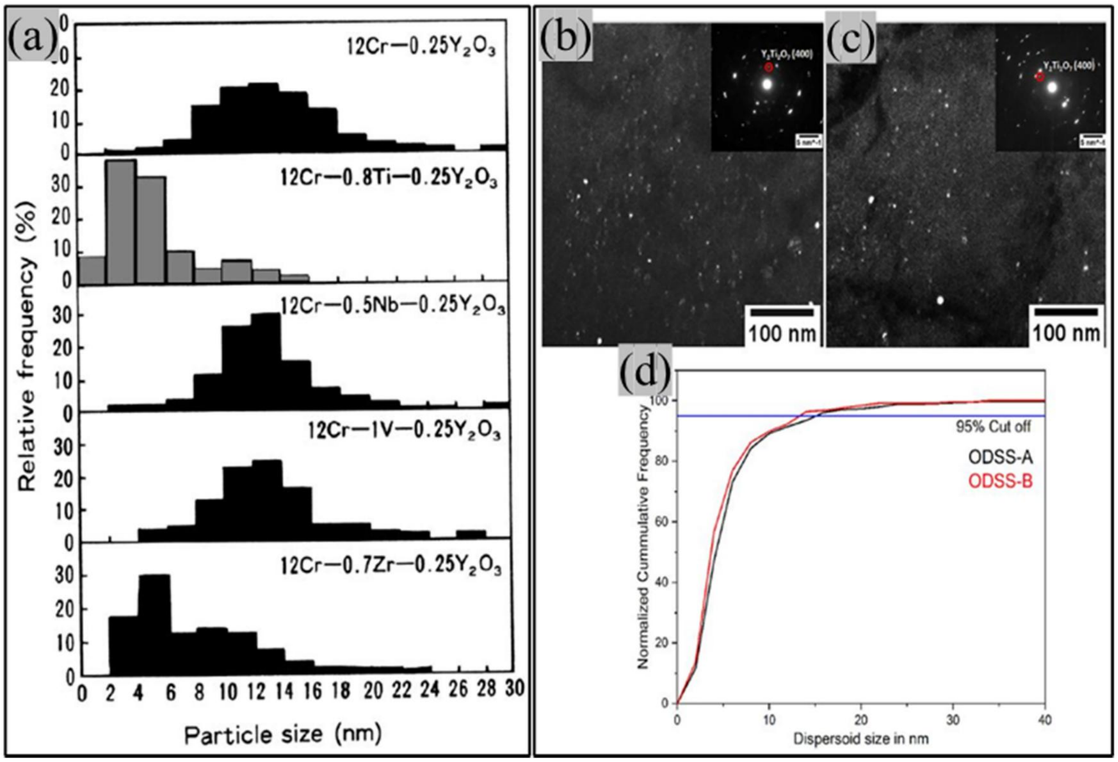


**Figure 15.** (a) Size distribution of the nano oxide particles on varying alloying elements with constant yttria content in ferritic ODS steels.[43] (b,c) Dark field TEM images of oxide particles and (d) corresponding their size distribution with different yttria content in F/M ODS steels.[84]

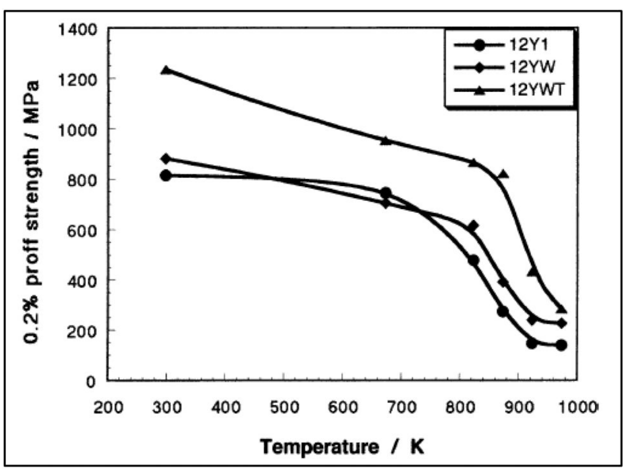


**Figure 16.** Effect of Ti and W on the yield strength of a 12Cr ODS steel at various temperatures.[99]

Baluc[136] have also observed that increasing the Ti content in the alloy reduces the mean particle size and increases the number density of the Y-Ti-O oxide particles. They have reported that Ti content should be below 0.3% in the ODS steels to avoid the formation of large (50–500 nm) $TiO_2$ particles, which have a detrimental effect on the mechanical properties of ODS steels. He et al.[139] have prepared alloys Fe-13.5Cr-2W-xTi-0.3$Y_2O_3$ with varying Ti concentrations ($x = 0$, 0.2, 0.3, 0.4). They found that 0.3%Ti ODS steels showed the finest dispersoid size and exhibited excellent tensile strength among the other Ti concentration values analyzed.

Yttria ($Y_2O_3$) is known to be one of the preferred dispersoids in ODS steels. Creep resistance can be enhanced by incorporating dispersoids, such as yttria ($Y_2O_3$), into the ferrite matrix through mechanical alloying. Yttria is preferred over nitrides and carbides due to its superior thermal stability at elevated temperatures. It provides better strength than other oxides such as $TiO_2$, $MgAl_2O_4$, and $ZrO_2$[39] and exhibits better stability under neutron irradiation than $MgAl_2O_4$ oxides.[140] In recent years, other rare earth oxides such as $CeO_2$, $La_2O_3$, $Gd_2O_3$, and $Er_2O_3$ are also being considered as potential additives in the production of ODS alloys.[141–143] The Gibbs free energy over temperature of various oxides is presented in Figure 17.

Hoffman et al.[141] have extensively studied the effect of various oxides in HIPed ODS steels with compositions Fe-13Cr-1W-0.3Ti + 0.3$M_xO_y$ ($M_xO_y$= MgO, $La_2O_3$, $Ce_2O_3$, $ZrO_2$, and $Y_2O_3$). They found that the yttria-containing alloys exhibited higher tensile strength at room temperature as well as at elevated temperature (Figure 18a). However, the alloys containing MgO exhibited better Charpy impact values than the other alloys when tested at temperatures ranging from −100 to 400 °C (Figure 18b). The higher strength is achieved through high temperature stability and uniform dispersion of fine yttria particles. Raghavendra et al.[144] studied the mechanical alloying of Fe with 15 wt.% $ZrO_2$ as a dispersoid. They found that even after 100 h of milling, $ZrO_2$ remained crystalline, unlike $Y_2O_3$, which tends to amorphize under similar conditions—an effect that can compromise structural stability.[145] This stability of $ZrO_2$ is due to its smaller number of atoms per unit cell and is considered beneficial. Since amorphization can cause unwanted nucleation and rapid growth of dispersoids during high-temperature consolidation, $ZrO_2$ was chosen as a suitable alternative. Based on this, they developed 9Cr ODS steels using $ZrO_2$ as the dispersoid.[146]

Many researchers have focused on ODS steels containing up to 0.35% $Y_2O_3$.[52,85,86,147] Studies on steels with higher yttria contents are limited because increased yttria levels can cause a drastic reduction in ductility. Few studies are carried out on ODS ferritic-martensitic steels (Eurofer97) containing higher yttria content.[148,149] Lucon studied the effect of yttria in HIPed Eurofer97 steel. He reported that increasing the yttria content from 0.3% to 0.5% led to an increase in the tensile strength,[149] accompanied by a slight reduction in ductility (from ∼30 to ∼27%). Cayron et al.[140] used up to 1% yttria in the Eurofer97 ODS steel consolidated through HIP. They obtained a slightly higher value of tensile strength ∼1168 MPa, but the alloy exhibited poor ductility of ∼0.8%. Ukai et al.[41,97] have extensively studied the effect of higher yttria content in F/M ODS as well as ferritic ODS steels. They added up to 0.94% yttria in extruded F/M ODS steels.[97] It has been observed that the room and high temperature (700 °C) tensile strength along with creep life increase with increasing the yttria content in the alloys. In extruded ferritic ODS steels with compositions Fe-13Cr-3W-0.5Ti-$xY_2O_3$ ($x = 0.22$, 0.30, 0.39, 0.48 and 0.56%). The increase in strength and decrease in ductility was observed on increasing the yttria content. However, the creep strength gets saturated up to 0.4% yttria contents, and a huge difference in uniaxial and biaxial creep strength is observed at 700 °C. They have also observed that on increasing the yttria content, the creep strength exhibits increased anisotropy. Increasing the yttria content leads to a higher number density of complex oxides, which effectively pins the elongated grains and prevents recrystallization. The effect of yttria content on tensile strength and ductility is shown in Figure 19a. The effect of yttria on uniaxial as well as biaxial creep strength is shown in Figure 19b.

Aluminum (Al) addition in ODS steels is found to be effective in improving corrosion resistance. It has been shown that high- Cr ODS steel containing more than 4% Al exhibited significantly higher corrosion

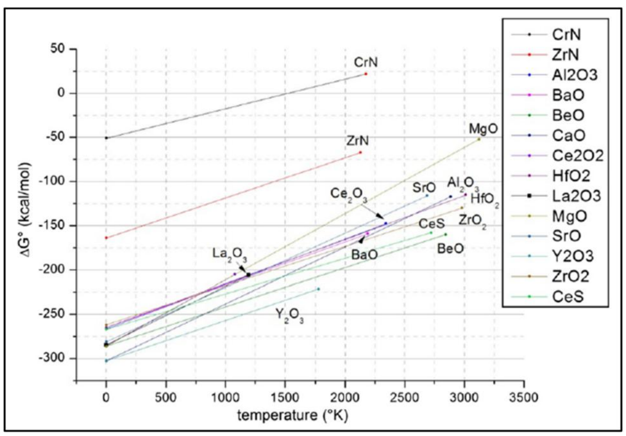


Figure 17. Gibbs-free energy vs. temperature for various oxides and nitrides.[141]

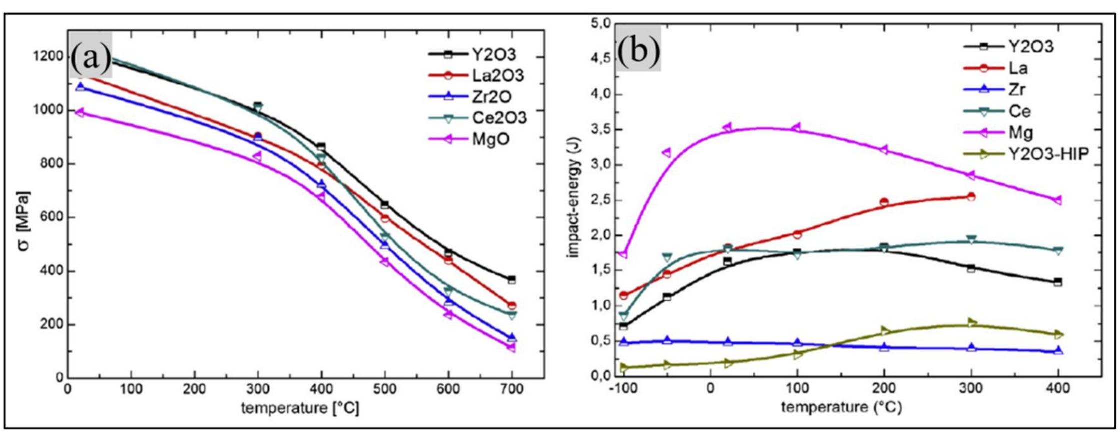


Figure 18. Effect of various oxides on (a) yield strength and (b) variation of Charpy impact energies of a Fe13Cr1W0.3Ti + 0.3 $M_xO_y$ (wt. %) steel at various temperatures.[141]

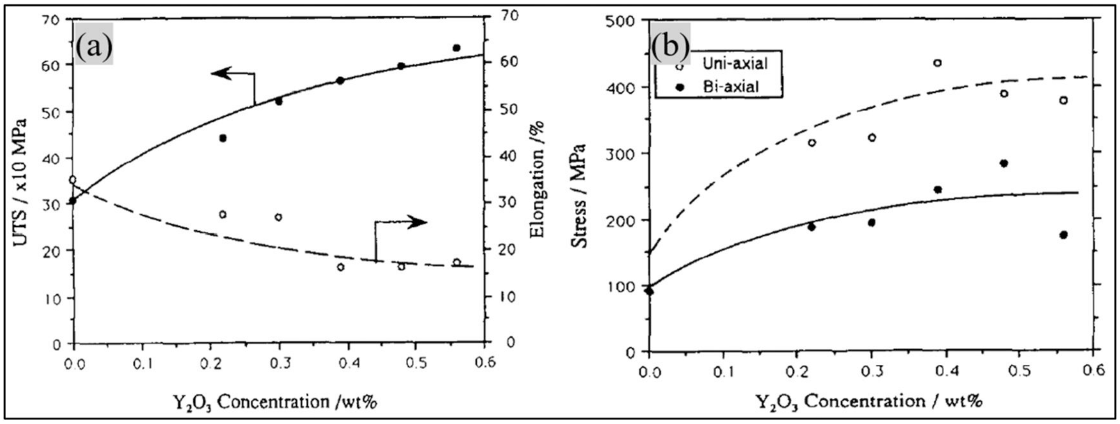


**Figure 19.** (a) Effect of yttria on ultimate tensile strength and elongation of 13Cr-3W-0.5Ti steel and (b) variation of its uniaxial and biaxial creep strength with yttria contents at 650 °C.[41]

resistance than the 9Cr ODS steels without Al.[150,151] Studies have shown that Al has a positive effect on the compatibility of materials in lead-bismuth eutectic (LBE) environments. Alloys containing 8–15 wt% Al exhibit excellent corrosion resistance at temperatures between 420 and 600 °C, primarily due to the formation of a protective alumina scale.[152] It is well known that 9Cr ODS steels exhibited relatively lower corrosion resistance than high Cr (>12%Cr) ODS steels. This limitation can be effectively overcome by adding Al, which enhances its protective properties. The addition of Al to ODS steels promotes the formation of a dense alumina ($Al_2O_3$) layer on the surface, providing effective protection against further corrosion of the underlying metal. However, Al can also influence the microstructure by forming coarse Y-Al-O oxide particles and reduce the high-temperature mechanical properties of the ODS steels.[153] Few other studies reported that the addition of Zr and Hf could suppress the formation of large Y-Al-O by forming fine Y-Zr-O and Y-Hf-O oxide particles, respectively.[154,155] These fine, complex oxide particles significantly enhance high-temperature strength while maintaining the corrosion resistance of the steels. Yu et al.[156] investigated the effects of Zr, Hf, and a combination of Zr + Hf in a Ni-based ODS superalloy containing Al. They studied how these alloying elements influence optimization of microstructure (including grain structure, gamma prime phases, and complex nano-oxides) and their role in enhancing the mechanical properties. They observed that the addition of Hf led to a more significant grain size refinement compared to the addition of Zr or the combination of Zr and Hf. However, neither Zr nor Hf had a noticeable effect on the size or volume fraction of the gamma prime ($\gamma'$) precipitates.

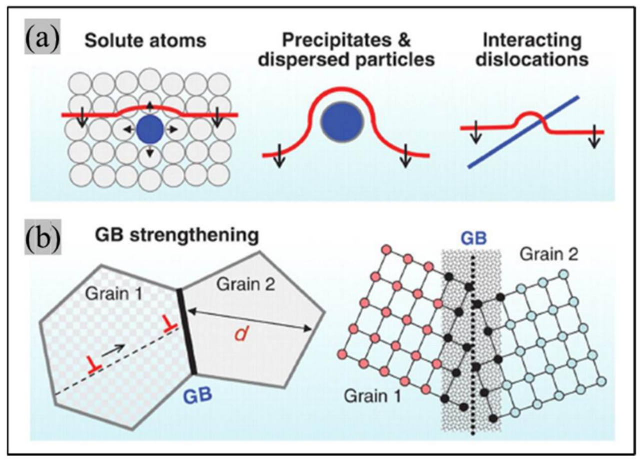


**Figure 20.** Schematic representation of (a) solid solution strengthening and (b) grain boundary strengthening mechanisms. The precipitate and dislocation interaction are represented in blue and red colors, respectively.[157]

## 8. Strengthening mechanism in ODS steels

Materials are strengthened by the formation of internal defects and boundaries, which impede the dislocation motion.[157] A combination of strengthening mechanisms may be operative in ODS steels. Figure 20a shows strengthening due to solid solution. The solute atoms create a lattice strain around them, which provides obstacles to dislocation motion. Sometimes dislocations interact with each other, which hinders their motion, leading to strengthening. This is called dislocation forest strengthening. The ODS alloys are mainly prepared through the powder metallurgy route. Mechanical alloying generates high dislocation densities in the alloys, which may not be fully relieved during consolidation. Hence, it contributes to the strengthening of ODS alloys. In grain boundary or Hall-Petch strengthening, dislocation motion is hindered by the grain boundaries due to an

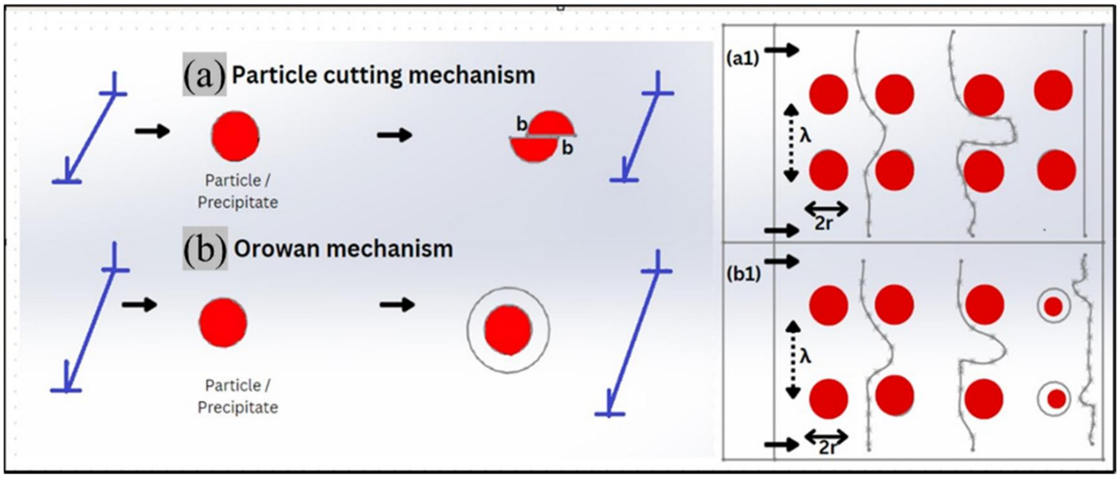


Figure 21. A schematic showing (a) particle cutting and (b) Orowan looping by the hard particles.[68] The corresponding figures ($a_1$) and ($b_1$) show steps of (a) particle shearing (cutting) by the dislocation and Orowan strengthening due to dislocation looping by hard particle (Adapted from [68] and [160]).

abrupt change of orientation of the slip plane from one grain to another one (Figure 20b). Higher stress is needed to transfer the slip across the grain boundary. Hence, polycrystalline materials with finer grains (more GBs) have higher strength than the coarser-grained (less GBs) ones. Oxide particles refine the grain structure during mechanical alloying and consolidation by pinning grain boundaries. This leads to very fine grain sizes (~150 nm to ~300 nm) in ODS alloys. Therefore, grain boundary strengthening also plays a significant role in the hardening of ODS alloys. Details of strengthening contributions from each mechanism are available in the literature.[68,104,158]

Based on specific microstructural features, several authors have reported different contributions toward strengthening. Schneibel et al.[159] reported that the contribution due to Hall-Petch strengthening is approximately the same or greater than dispersion strengthening in the ODS PM2000 and NCS 14YWT steels. Similarly, Kim et al.[158] also found that contribution due to grain boundary strengthening is highest among others in 14YWT ODS steel. It is also important to note that the contribution of strengthening changes significantly with temperature. Alinger et al.[160] mentioned that the absolute melting temperature ($T_m$) of the materials affects the effectiveness of the strengthening mechanism. They have reported that precipitation or dispersion strengthening is more effective at a temperature greater than $0.5\,T_m$ (reactor temperatures at $0.5\,T_m$–$0.7\,T_m$), as the strengthening effect due to work hardening would be annealed out at relatively low temperatures.

Precipitation is a specific phase transformation in which the solute elements precipitate out from the supersaturated solid solution on lowering the temperature. Depending on their size, volume fraction, strength, and coherency, precipitates impede dislocation motion, which enhances the strength. The main drawback of precipitation strengthening is that it becomes less effective at high temperatures for extended periods of time. On exposure to elevated temperatures and prolonged times, precipitates generally coarsen through Ostwald ripening, which increases the interparticle distance and decreases the strength.[160] The precipitates eventually become soluble in the matrix at extremely high temperatures, leading to a loss in strengthening. The principle of dispersion strengthening involves introducing insoluble second phases into the ODS alloy, which remain stable up to the matrix's melting point. While dispersing stable oxide particles into the matrix addresses many of the drawbacks associated with precipitation strengthening, it also presents certain challenges.[66,67]

### 8.1. Interaction of particles and dislocations (dispersion strengthening)

When dislocations interact with oxide particles, the oxide particles can act in two ways to slow down dislocation motion. Either the dislocations cut the particles (particles-cutting mechanism) or bend around to bypass them (Orowan mechanism). These two mechanisms are represented in Figure 21. The interaction between particles and dislocations is highly influenced by the particle's strength. If the second-phase particles are hard enough, then shearing of particles by the dislocation may not be possible, and the dislocations will then have to bend around to bypass the particles. This bending around the non-shearable particles requires a larger stress and provides a significant strengthening effect.

One of the important factors in the dispersion strengthening is the interparticle distance ($\lambda$) between the oxide particles. The value of $\lambda$ mainly depends on the size (radius, $r_p$) and volume fraction ($V_f$) of the oxide particles. The mean interparticle distance is given by the following equation[160]:

$$\lambda = \sqrt{\frac{2\pi}{3V_f}}\, r_p \tag{1}$$

The increase of critical resolved stress due to dispersion of oxide particles over the alloy without dispersion is given by the equation[160]:

$$\Delta\tau_{\mathrm{crs}} = \alpha\mu b/\lambda \tag{2}$$

Where $\alpha$ is the barrier strength, $\mu$ is the shear modulus, and $b$ is the Burgers vector of the material. When the obstacles are non-shearable, the value of $\alpha$ is 1. This is known as Orowan strengthening. The Orowan looping and particle shearing (cutting) by dislocation mechanisms are shown in Figure 21. From Equation (2), it can be seen that an increase in strength is inversely proportional to the interparticle distance. For a constant volume fraction, particle refinement will lead to a decreased $\lambda$, which results in an increase in strength.

The classical Orowan stress, as established by Martin,[161] is defined as the stress required for dislocations to bypass precipitates by bending around the particles and looping, in the absence of thermal activation. This stress is given by the following equation[162]:

$$\sigma_{Or} = \frac{0.81\ M\ \mu\ b}{2\pi\sqrt{1-\nu}(\lambda - 2r_s)} \cdot ln\left(\frac{r_s}{b}\right) \tag{3}$$

Where $\sigma_{Or}$ is the Orowan stress (in MPa) required to bypass the particles; $M$ is the Taylor constant (3.06); $\mu$ is the temperature-dependent shear modulus of the material, calculated using the established equation $\mu = 89.5[1-0.84((\mathrm{T}\text{-}300)/\mathrm{T_m})]$, with $\mathrm{T_m}$ being the melting temperature of the material in K. Additionally, $b$ is the Burgers vector, and $\nu$ is the Poisson's ratio of the material.

## 9. Mechanical response of ODS steels

Several studies related to mechanical testing have been performed on ODS steels. The majority of them focused on determining their monotonous behavior, such as tensile, creep, and fracture toughness. Studies on fatigue and creep of ODS steels are limited.

### 9.1. Tensile behavior

ODS steels offer superior tensile strength compared to the base materials. ODS steels containing Ti form fine complex Y-Ti-O ($Y_2TiO_5$ and/or $Y_2Ti_2O_7$) oxide particles and exhibit higher tensile strength than ODS steels containing only pure coarse $Y_2O_3$ oxide particles.[113] The alloys having a very fine (2 nm − 10 nm) distribution of oxide particles along with fine-grained microstructure and high dislocation densities exhibited high strength. However, increasing strength in ODS steels typically compromises their ductility and fracture toughness.[163,164] F/M ODS steels exhibit higher yield and ultimate tensile strength but have a significantly lower value of ductility than ferritic steels due to the presence of martensite.[38] Their toughness can be improved by tempering. In such cases, there is a compromise on the strength of ODS steels. Ukai et al.[43] have extensively studied the tensile behavior of different F/M ODS steel compositions. These steels exhibited higher ultimate tensile strength (UTS) and uniform elongation (UE) than the conventional F/M steels, which do not have any oxide dispersion. Furthermore, they have observed that improvement in strength and ductility is more prominent at temperatures above 600 °C due to pinning of dislocations by the very fine distribution of nano oxide particles. However, Schaublin et al.[164] have shown that, at room temperature, the yield strength of F/M ODS steel containing only yttria is significantly higher than F/M ODS steel containing both yttria and titanium. On increasing the temperature, this difference decreases, and at 600 °C these values become equal. Kumar et al.[118] studied the tensile behavior of different ferritic ODS steel compositions. They have observed that both yield and ultimate tensile strength increase with increasing the yttria content. However, this was accompanied by a loss in ductility. Sundararajan et al.[165] extensively studied tensile properties of both F/M ODS (9%Cr) and ferritic ODS (18%Cr) steels. They have shown that 9Cr ODS steel shows higher tensile strength than 18Cr ODS steel up to 400 °C. At higher temperatures, the rate of decrease in tensile strength of 18Cr ODS steel is lower than that of 9Cr ODS steel. In both steels, strength and ductility decrease with increasing temperature. Moreover, it has also been observed that tensile properties are sensitive to the strain rate at high temperatures. Kumar et al.[117] have shown that, at 700 °C, both strength and ductility increase with increasing applied strain rates. A higher strain rate allows less time for recovery and strength increases due to work hardening. Ductility also increases as the time for cavitation to occur decreases. Cavitation is a major cause of failure at high temperatures. Similar results were also observed by other investigators.[166,167] Ferritic ODS steels (>12%Cr) show more anisotropy in mechanical properties than F/M ODS steels (containing <12%Cr).[41–43]

Recently, Xu et al. have demonstrated that novel heterostructure ferritic ODS steels possess exceptional mechanical properties, exhibiting a yield strength of approximately 2 GPa and a ductility of ~13%.[168] These steels are developed using mechanical alloying followed by a two-step consolidation process involving high-pressure sintering and subsequent hot rolling. The developed heterostructure ferritic ODS steels exhibited microcrystalline (MC) domains (~35% volume fraction) dispersed within an ultrafine-grained (UFG) matrix (~500 nm grain size). This unique microstructure enables the alloy to achieve an ultra-high yield strength along with excellent tensile elongation. Similarly, Cai et al.[169] synthesized novel bulk nanocrystalline 14YWT ferritic ODS alloys by incorporating 0.8 wt% Zr through mechanical alloying followed by high-pressure consolidation. These alloys were developed through a collaborative effort between Oak Ridge National Laboratory (ORNL), the University of California, Santa Barbara, and Los Alamos National Laboratory (LANL), with the goal of achieving an optimal balance of strength, toughness, and irradiation tolerance.[170]

### 9.2. Impact behavior

ODS steels exhibit superior tensile strength, but inferior Charpy impact properties compared to conventional steels.[107,171–173] The presence of yttria in ODS steels decreases the toughness, which also leads to an increase in the ductile-to-brittle transition temperature (DBTT).[149] It has been observed that the milling atmosphere may also affect the impact properties of the ODS steels. Oksiuta et al.[107] have shown that the alloys milled in a hydrogen atmosphere exhibited higher impact energy than the same alloy compositions milled in an argon atmosphere. The consolidation route for these steels also greatly affects their impact energies. Ukai et al.[43] have observed that extruded ODS steels exhibited high impact energy. However, HIPed ODS steels exhibited extremely low values of impact energy that could be attributed to the presence of pores or precipitates at the grain boundaries of the sintered particles. By applying various thermo-mechanical treatments (such as rolling and forging), a significant improvement in the impact energy can be achieved in the HIPed ODS steels. This improvement is due to the absence of porosity by applying plastic deformation through rolling and forging. Similar to tensile samples, Charpy samples exhibit strong anisotropy. Rouffié et al.[174] have shown strong anisotropy in the impact energy of the extruded samples. The samples prepared along the extrusion direction exhibited higher impact energy than the samples prepared along the transverse direction. Crystallographic fiber texture <110> along the extrusion direction leads to anisotropy in extruded alloys.[175] Fournier et al.[176] have prepared samples along different orientations (i.e. longitudinal (LT), transversal (TL), and 45°) from the extruded rod. The LT samples exhibited higher impact energy (Figure 22). Hadraba et al.[175] have also observed a similar variation in impact behavior.

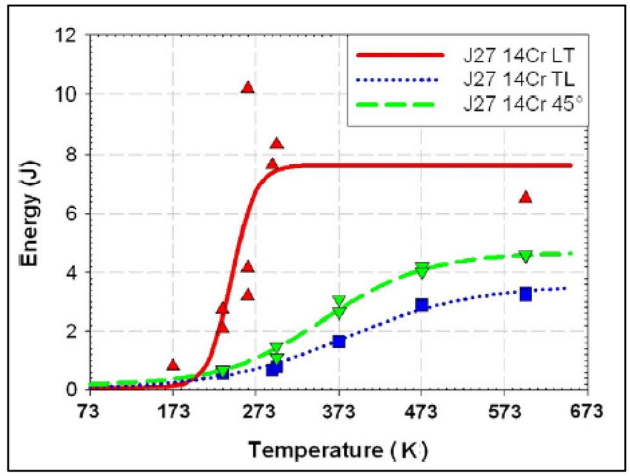

Figure 22. Anisotropy in the Charpy impact behavior at different sample orientations of extruded ferritic ODS steel J27.[176]

### 9.3. Creep behavior

Ferritic steels exhibit BCC crystal structure, which has poor creep resistance because its open structure leads to higher diffusion rates and dislocation mobility at elevated temperatures. Huet et al.[177] successfully and significantly enhanced the creep resistance of these steels by incorporating oxide dispersions into their microstructure. The behavior of conventional 18Cr ferritic steel and 18Cr ODS ferritic steel is represented in Figure 23a.

Creep deformation of materials depends on various factors such as composition, test temperature, applied stress, grain size, and fraction of second-phase

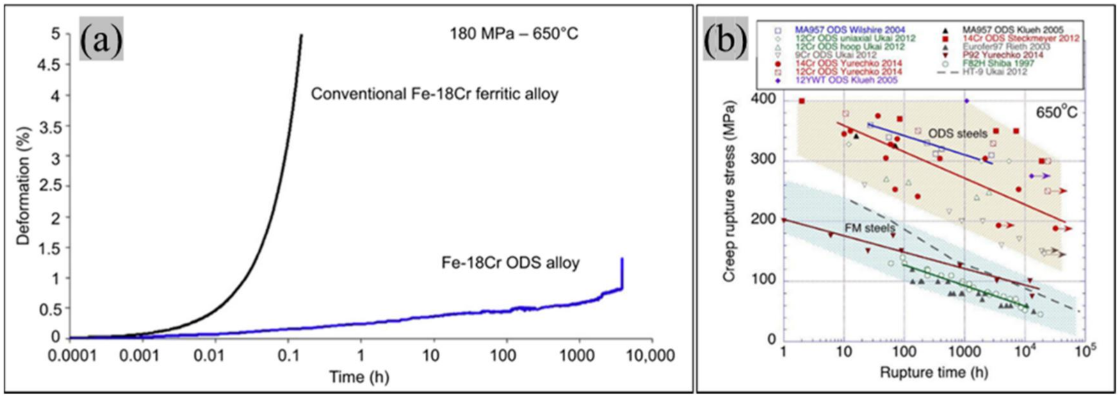


**Figure 23.** (a) Comparison in creep behavior of 18Cr ferritic steel with 18Cr ferritic ODS steel at 180 MPa and 650 °C[38] and (b) creep behavior of various ODS steels at 650 °C.[178]

particles. Klueh et al.[179] studied the effect of titanium (Ti) addition on the creep behavior of 12Cr ODS steel. The addition of titanium significantly improves the creep life of ODS steels. Ukai et al.[41] have investigated the effect of yttria and titanium addition on uniaxial and biaxial creep behavior of 13Cr ODS steels. Both the uniaxial and biaxial creep life increase on increasing yttria content. The optimum yttria content mentioned for Fe-13Cr-3W-0.5Ti ODS steel was 0.4 (wt.%). Hayashi et al.[180] evaluated the creep life of 14Cr ODS steel annealed at 940 °C for different aging time lengths. Grain growth during creep was not significant enough to affect the creep life of these steels. Steckmeyer et al.[91] showed strong anisotropy in the creep behavior of 14CrWTi ODS steel produced by extrusion. Malaplate et al.[181] have observed that the creep life of 18Cr ODS steel was higher than that of MA957 alloy. The comparison between creep life of conventional steels and ODS steels in the literature is summarized in Figure 23b. However, the detailed creep mechanism for these ODS alloys was left undiscussed in most of the literature. Recently, Jarugula et al.[162] have investigated the creep mechanism over the temperature range of 650–750 °C and stress range of 175–400 MPa in extruded 18Cr ODS steels. They found that dislocations climb over the particles, and dislocation detachment on the departure side of particles is the main creep mechanism operating in these steels.

## 10. Current trends in ODS steels for next generation nuclear power plants

Most studies have focused on ferritic-martensitic (F/M) ODS steels with chromium (Cr) content ranging from 9 to 11 wt.%.[43,73,182] These steels exhibit phase transformation above 800–850 °C, which limits their use at extremely high temperatures.[37] They also exhibit lower corrosion resistance than high-chromium ferritic ODS steels. However, recycling spent fuel is one of the main requirements in fission nuclear power plants. Filtration of unspent fuel might be extremely challenging due to the partial dissolution of clad tubes in boiling nitric acid medium during the initial steps of the solvent extraction method.[46,102,183] Due to this, high-Cr ODS fuel-clad tubes should be developed, which would increase their resistance to dissolution during reprocessing.[46,183] Ferritic steels having low chromium (<14%) content may not meet the requirements of fast breeder reactors (FBRs). Ferritic ODS steels with higher chromium content (18 wt.%) are being developed for Gen-IV nuclear reactors due to their excellent corrosion and oxidation resistance. These steels do not undergo a ferrite-to-austenite transformation and maintain the ferrite phase up to the melting point. Limited literature is available on 18Cr ferritic ODS steels,[52,86,102] as most of the work on ferritic ODS steels is done for compositions between 12 and 16% Cr content.[48–50,184] These steels are conventionally made by extrusion of ODS steel powders. The extruded alloys exhibit anisotropy, which increases with increasing Cr content.[37] Besides excellent swelling resistance, ferritic ODS steels also exhibit very good strength and ductility at room temperature and creep strength at a temperature of 650 °C and above.[38] High-chromium ODS steel maintains its integrity during spent fuel reprocessing and is a viable choice for Gen-IV nuclear reactors.[54] Therefore, high Cr (maximum 18% Cr) is being developed for these nuclear power plants.

Nagini et al.[135] studied the corrosion behavior of 18Cr ODS steels and found high corrosion resistance, suggesting that the use of 18Cr ODS steels is beneficial for their use in environments where corrosion is a concern. Deepak et al. developed 18Cr ODS steels through powder forging and studied the effect of yttria content on mechanical behavior. They found that higher yttria-containing alloys exhibited high tensile strength but at the expense of ductility. Ratnakar et al. studied the tensile and creep behavior of 18Cr ODS steels and concluded that these steels exhibit higher tensile and creep properties compared to low Cr-containing ODS steels. This suggests that 18Cr ODS steels may have superior mechanical properties, particularly in terms of tensile strength and resistance to corrosion and creep deformation. Further comprehensive investigation is warranted to thoroughly examine the mechanical and corrosion behavior of these steels, particularly in the context of their potential application as clad tube materials for upcoming Generation IV nuclear reactors.

## 11. Summary

ODS steels are primarily being developed as a clad tube material for Gen-IV nuclear reactors. Clad tubes separate fuels from the coolant and experience high temperature and radiation. It also experiences a biaxial state of stress. Hence, the isotropic properties of ODS steels will play a very important role in Gen-IV nuclear reactors. The clad tube having bands/stringers may fail in a transverse direction. The presence of these bands/stringers is undesired because creep cavitation may preferentially take place along these bands. These bands/stringers can be avoided by selecting suitable consolidation routes to fabricate ODS steels. Therefore, consolidation through the hot powder forging route of these steels can improve the life of the power plant. Powder-forged steels exhibit a recrystallized microstructure that produces isotropic mechanical properties. By using powder forging instead of extrusion, anisotropy in high-Cr steels can be reduced. The powder forging process produces alloys with more isotropic mechanical properties, as it avoids the formation of prior particle boundaries (PPBs) and extrusion bands (stringers), leading to improved material characteristics. The present work extensively reviewed the ODS steels and their fabrication routes. This work also identified the potential candidate materials being developed for Gen-IV nuclear reactors. From the review of available literature, the following points summarize the present study:

1. ODS steel studies are done using pre-alloyed powders or elemental powders; the use of elemental powders offers flexibility in the choice of composition.
2. Conventional consolidation techniques used to produce ODS steels are hot extrusion (HE) or hot isostatic pressing (HIP). However, extruded material suffers from anisotropy, while HIPed material exhibits PPBs. Hence, to achieve isotropic mechanical properties without any PPBs, a new consolidation route, i.e., hot powder forging, may be used as a viable alternative to consolidate ODS steel powders.
3. Extruded ODS alloy exhibits stringers/bands parallel to the extrusion direction. These bands are detrimental to creep properties. No band formation is expected in hot powder forging, as material flow is not unidirectional.
4. Extruded ferritic ODS steels (>12Cr (wt. %)) show anisotropy in the mechanical properties compared to the F/M ODS steels (<12%Cr (wt. %)), and there is an anisotropy increase in the mechanical properties with increasing Cr content. Powder forging in consolidation of ODS steel powders can overcome these problems.
5. It is very difficult to achieve a fully recrystallized microstructure in extruded ferritic ODS steel containing more than ∼0.27 yttria (wt. %). However, consolidation through hot powder forging may produce recrystallized microstructure in ODS steels with high yttria contents (up to 1 wt. %).
6. In most ODS steels, yttria content has been restricted to 0.35 wt.% because higher concentrations lead to a significant reduction in ductility. However, this limitation has been overcome using powder forging for consolidating ODS steel powders, allowing yttria content to be increased up to 1 wt.% in ferritic 18Cr ODS steel.
7. Hot powder forging takes less time to consolidate ODS steel powders than conventional techniques, leading to the development of a high-density product with fine-grained recrystallized microstructure, satisfactory ductility, and comparable mechanical properties to those produced via hot isostatic pressing and extrusion.

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